

Cubes & Dices

Are you preparing for SSC CGLNOYES

Here, '=' represents Edges

'→' represents Surfaces

and, '●' represents Corners

In exams two or more positions of a dice are given in a question and your task is to find out the opposite surface of a given surface.

Example:

From the above two positions of a dice it is clear that 2,3,4 and 5 are the adjacent surfaces of surface '1'. Therefore, we can say that opposite surface of '1' is surface '6'.

TYPES OF DICE

1. **STANDARD DICE:** The dice in which the sum of two opposite surfaces is equal to seven is considered as Standard Dice.

For Example:

In the above dice, 5,3 and 1 are adjacent surfaces and it is clear that the sum of any two surfaces shown in dice is not equal to seven.

Therefore, Surface opposite to '5' is Surface '2' (i.e $5+2=7$)

Surface opposite to '3' is Surface '4' (i.e $3+4=7$)

Surface opposite to '1' is Surface '6' (i.e $1+6=7$)

1. **GENERAL DICE:** In General Dice the sum of two opposite surfaces is not equal to seven. To find out the opposite surface we are required atleast two positions of dice having:

(a) one surface common

Here, Surface '3' is common surface in both the positions of dice. After fixing '3' and then moving in clockwise direction we can easily find the opposite surfaces.

3 – 6 – 4

3 – 5 – 2

It is clear that 6 and 5 are opposite surfaces and, 4 and 2 are opposite surfaces. Therefore, surface opposite of surface '3' is surface '1'.

(b) Two surface common

In the above two positions of dice, two surfaces i.e Surface '3' and Surface '2' are common.

3 – 2 – 1

3 – 2 – 6

After fixing surfaces 3 and 2, we find that surface opposite of surface '1' is surface '6'.

Surface opposite of surface '3' is not confirm, it may be surface '4' or surface '5'. Similarly, surface opposite of surface '2' is not confirm, it may be surface '5' or surface '4'.

III. OPEN DICE: In open dice all the six surfaces are clearly shown. The opposite surfaces are at the alternate positions of rows or columns. For Example:

Here, opposite of surface '1' is surface '4'

opposite of surface '3' is surface '5'

opposite of surface '2' is surface '6'

- **EXPANSION OF DICE:** In expansion of dice we are given two or more positions of dice and we have to represent all the six surfaces in open dice.

For Example:

In the above two positions of dice common surface is '4'.

Always fix the common surface at the centre position.

From first position:

i.e

From second position:

i.e

Clearly blank surface will be filled by remaining number i.e '6'

Therefore, opposite surfaces are: $1 \leftrightarrow 2$, $4 \leftrightarrow 6$ and $3 \leftrightarrow 5$

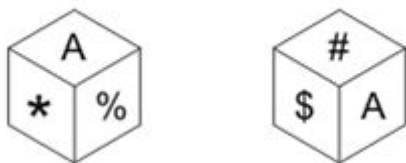
Cube and Dice

1. Which of the following cubes cannot be formed, when the given figure is folded to form a cube?



- a) D
- b) C
- c) B
- d) A

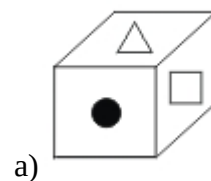
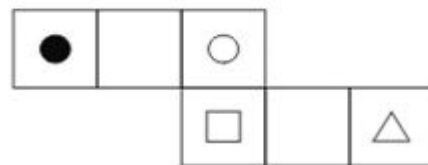
2. Two different positions of the same dice are shown. Select the symbol that will be on the face opposite to the one showing '#’.



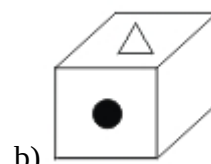
- a) %
- b) &
- c) *

d) %

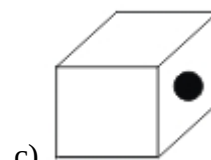
3. Which of the following cubes can be formed, when the given figure is folded to form a cube?



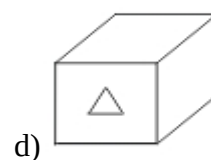
a)



b)

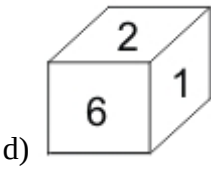
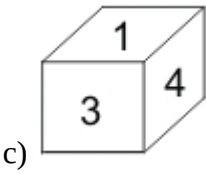
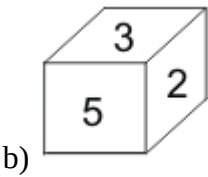
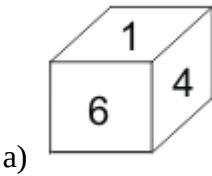
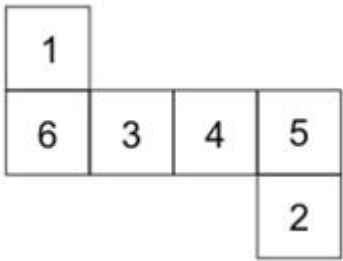


c)

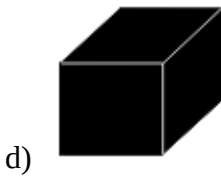
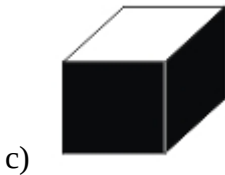
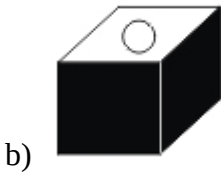
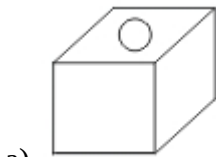
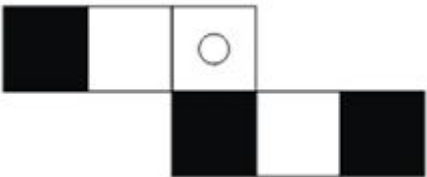


d)

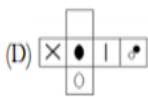
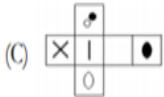
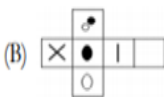
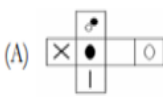
4. Which of the following cubes can be formed, when the given figure is folded to form a cube?



5. Which of the following cubes can be formed, when the given figure is folded to form a cube?



6. Choose the alternative that best resembles the unfolded form of the given cube.



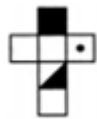
a) B

b) C

c) D

d) A

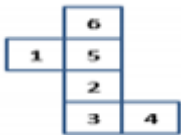
7. Which of the following cubes can be formed, when the given figure is folded to form a cube?



- (A)
- (B)
- (C)

- a) B
- b) C
- c) D
- d) A

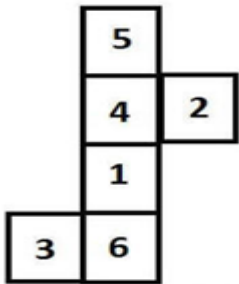
8. Which of the following cubes can be formed, when the given figure is folded to form a cube?



- (A)
- (B)
- (C)
- (D)

- a) B
- b) C
- c) D
- d) A

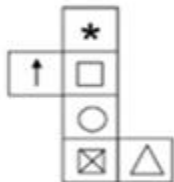
9. Which of the following cubes can be formed, when the given figure is folded to form a cube?



- A
- B
- C
- D

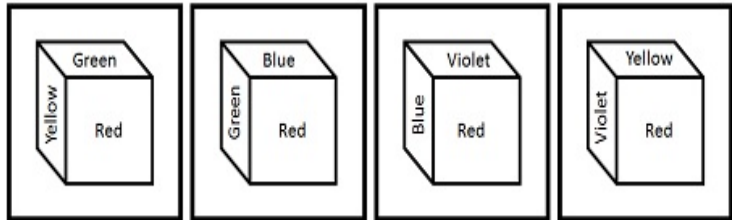
- a) B
- b) C
- c) D
- d) A

10. Which of the following cubes can be formed, when the given figure is folded to form a cube?



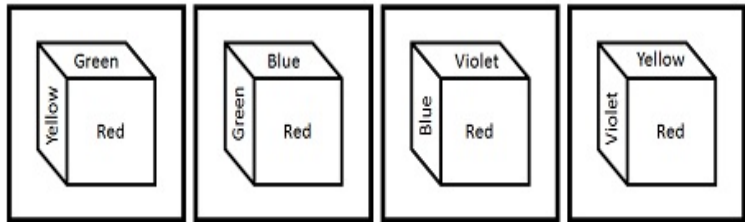
- a)
- b)
- c)
- d)

11. Four different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘Blue’.



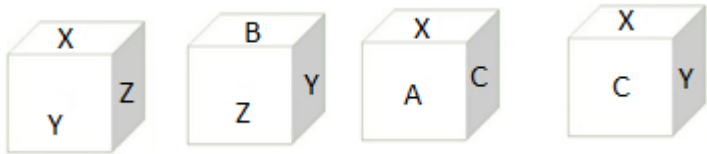
- a) Green
- b) Yellow
- c) Red
- d) Violet

12. Four different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘Violet’.



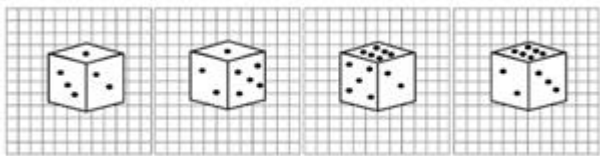
- a) Green
- b) Yellow
- c) Red
- d) Blue

13. Four different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘X’.



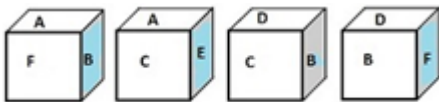
- a) Y
- b) B
- c) A
- d) C

14. Four different positions of the same dice are shown. Select the number of dots that will be on the face opposite to the one showing ‘4 dots’.



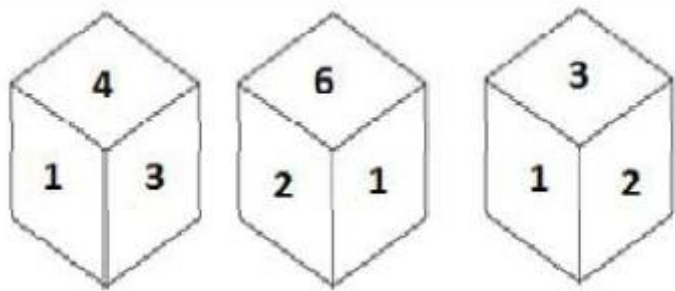
- a) 1
- b) 3
- c) 2
- d) 5

15. Four different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘C’.



- a) F
- b) A
- c) B
- d) E

16. Three different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing ‘4’.



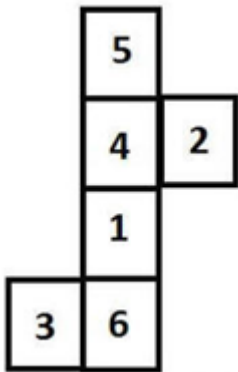
- a) 1
- b) 2
- c) 3
- d) 6

17. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘2’.



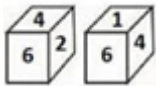
- a) 3
- b) 6
- c) 4
- d) 5

18. When the given figure is folded to form a cube, which number will not be adjacent to ‘2’?



- a) 5
- b) 3
- c) 1
- d) 6

19. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing ‘1’.



- a) 6
- b) 2
- c) 4
- d) 5

20. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing ‘3’.



- a) 1

- b) 2
- c) 5
- d) 6

21. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing '4'.



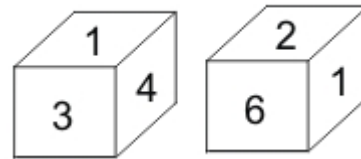
- a) 5
- b) 1
- c) 2
- d) 6

22. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing '4'.



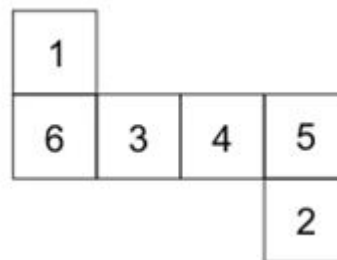
- a) 5
- b) 3
- c) 2
- d) 6

23. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing '6'.



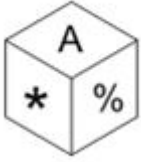
- a) 1
- b) 2
- c) 3
- d) 4

24. Select the correct combination of opposite faces, when the given figure is folded to form a cube?



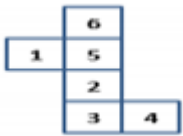
- a) 6, 1
- b) 5, 4
- c) 1, 3
- d) 4, 6

25. Two different positions of the same dice are shown. Select the symbol that will be on the face opposite to the one showing 'A'.



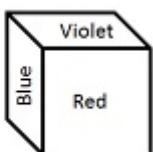
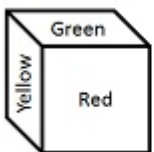
- a) %
- b) \$
- c) @
- d) %

26. Select the correct combination of opposite faces, when the given figure is folded to form a cube?



- a) 1, 6
- b) 5, 2
- c) 3, 4
- d) 5, 3

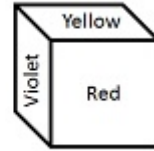
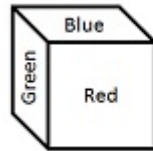
27. Two different positions of the same dice are shown. Select the colour that will be on the face opposite to the one showing 'Red'.



- a) Blue

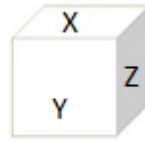
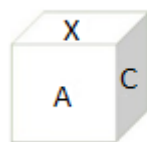
- b) Red
- c) Green
- d) Pink

28. Two different positions of the same dice are shown. Select the colour that will be on the face opposite to the one showing 'Violet'.



- a) Blue
- b) Red
- c) Green
- d) Yellow

29. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing 'Z'.



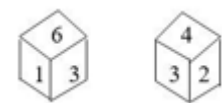
- a) B
- b) C
- c) X
- d) Y

30. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing ‘2’.



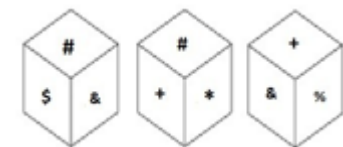
- a) 3
- b) 6
- c) 1
- d) 5

31. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘3’.



- a) 5
- b) 1
- c) 4
- d) 2

32. Three different positions of the same dice are shown. Select the symbol that will be on the face opposite to the one showing ‘*’.



- a) \$
- b) &

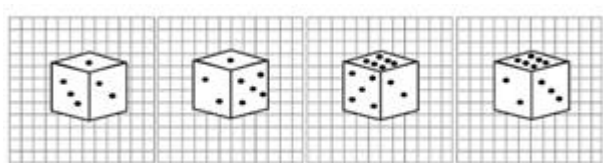
- c) %
- d) +

33. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘B’.



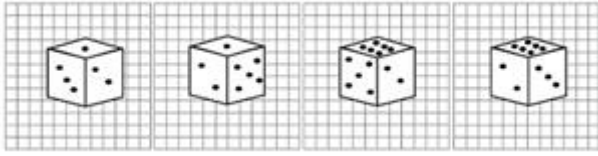
- a) X
- b) Y
- c) Z
- d) C

34. Four different positions of the same dice are shown. Select the number of dots that will be on the face opposite to the one showing ‘four dots’.



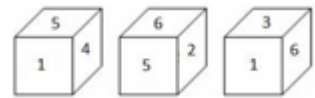
- a) 1
- b) 3
- c) 2
- d) 5

35. Four different positions of the same dice are shown. Select the number of dots that will be on the face opposite to the one showing ‘Three dots’.



- a) 1
- b) 3
- c) 2
- d) 5

36. Three different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing '5'.



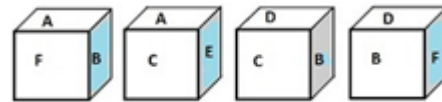
- a) 1
- b) 2
- c) 6
- d) 3

37. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing '6'.



- a) 5
- b) 3
- c) 1
- d) 4

38. Four different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing 'A'.



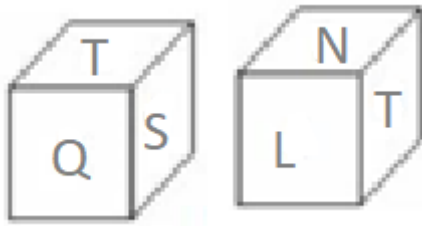
- a) F
- b) D
- c) B
- d) E

39. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing 'N'.



- a) O
- b) W
- c) E
- d) M

40. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing 'Q'.



- a) N
- b) L
- c) S
- d) T

41. Two different positions of the same dice are shown. Select the symbol that will be on the face opposite to the one showing ‘%’.



- a) @
- b) \$
- c) *
- d) !

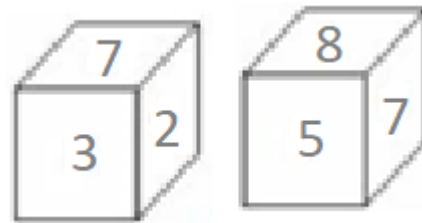
42. Two different positions of the same dice are shown. Select the letters that will be on the face opposite to the one showing ‘bl’.



- a) in

- b) ln
- c) pk
- d) kt

43. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing ‘8’.



- a) 3
- b) 2
- c) 8
- d) 5

44. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘M’.



- a) W
- b) O
- c) E
- d) N

45. Two different positions of the same dice are shown. Select the letters that will be on the face opposite to the one showing ‘kt’.



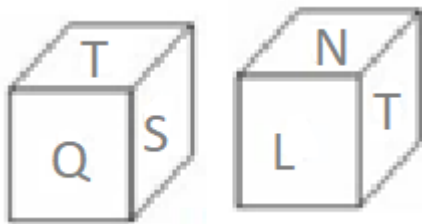
- a) pk
- b) bl
- c) ew
- d) ln

46. Two different positions of the same dice are shown. Select the symbol that will be on the face opposite to the one showing ‘*’.



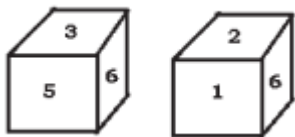
- a) !
- b) @
- c) %
- d) \$

47. Two different positions of the same dice are shown. Select the letter that will be on the face opposite to the one showing ‘S’.



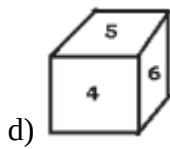
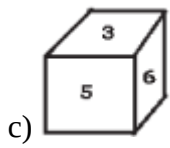
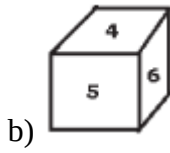
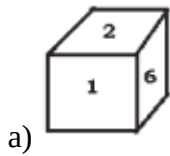
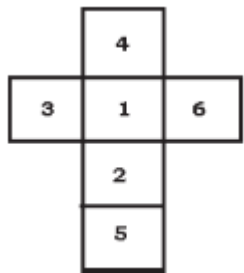
- a) T
- b) Q
- c) L
- d) B

48. Two different positions of the same dice are shown. Select the number that will be on the face opposite to the one showing ‘5’.

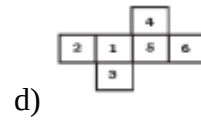
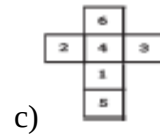
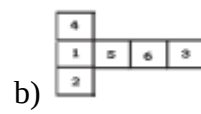
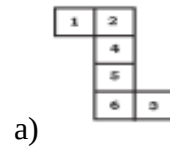


- a) 2
- b) 1
- c) 6
- d) 3

49. Which of the following cubes cannot be formed, when the given figure is folded to form a cube?

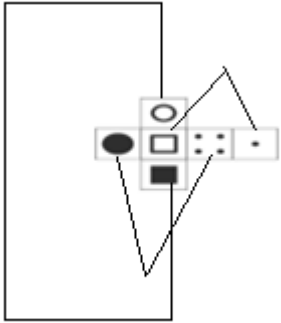


50. A standard dice is the one in which the sum of the numbers on the opposite surfaces equals to 7. Which of the below figures, when folded, can create a standard dice?



ANSWERS

1) Answer: A



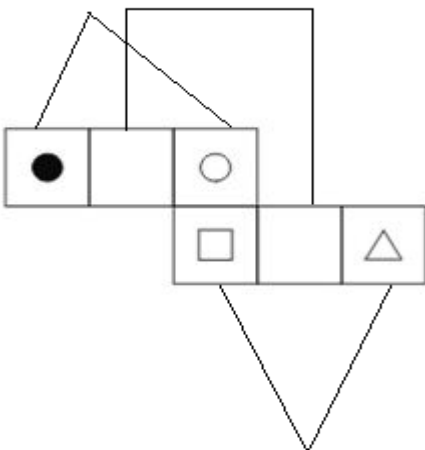
In option 'd', two opposite faces appear together. So, the cube cannot be formed.

2) Answer: C

As, 'A' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

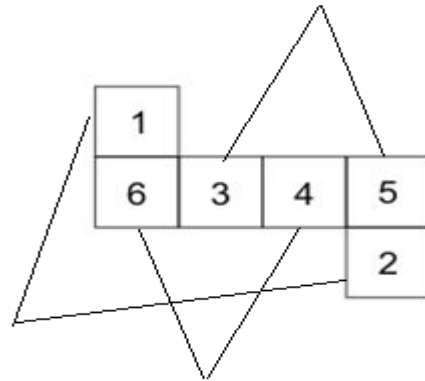
A	%	*
A	\$	#

3) Answer: B



In option 'b', no two opposite faces appear together, while in other three options, two opposite faces appear together. So, cubes other than 'b' cannot be formed.

4) Answer: C



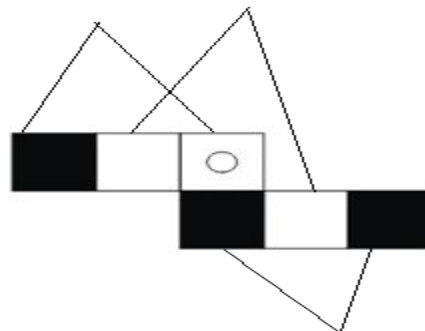
The faces opposite to each other in the dice is shown below:

6	3	2
4	5	1

In option 'c', no two opposite faces appear together. So, the cube can be formed.

In options other than c, two opposite faces appear together. So, the cubes cannot be formed.

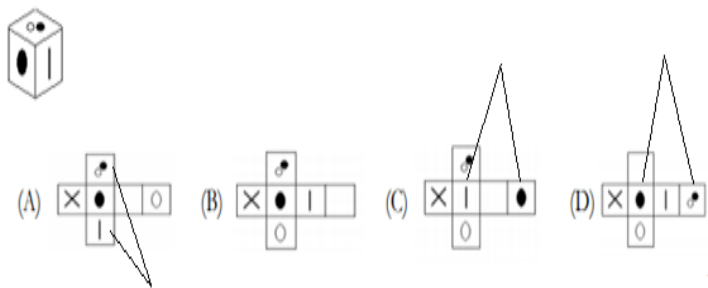
5) Answer: C



In option 'c', no two opposite faces appear together. So, the cube can be formed.

In options other than c, two opposite faces appear together. So, the cubes cannot be formed.

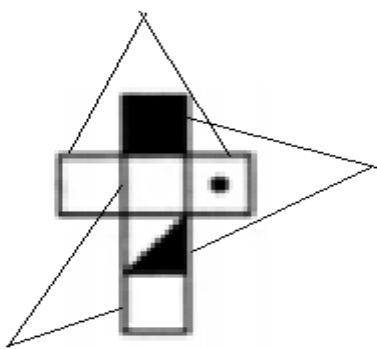
6) Answer: A



In options other than b, two opposite faces appear together. So, the cubes cannot be formed.

In option ‘b’, no two opposite faces appear together. So, the cube can be formed.

7) Answer: A



In option ‘b’, no two opposite faces appear together. So, the cube can be formed.

In options other than b, two opposite faces appear together. So, the cubes cannot be formed.

8) Answer: C

The faces opposite to each other in the dice is shown below:

6	5	1
2	3	4

In option ‘d’, no two opposite faces appear together. So, the cube can be formed.

In options other than d, two opposite faces appear together. So, the cubes cannot be formed.

9) Answer: D

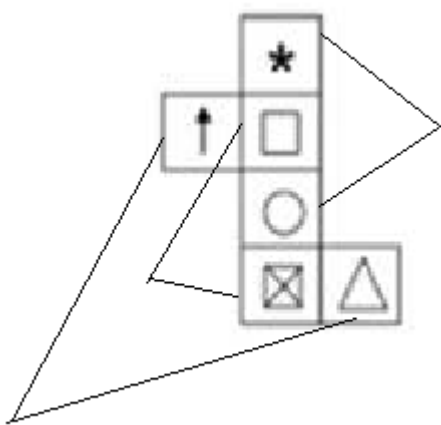
The faces opposite to each other in the dice is shown below:

6	5	3
4	1	2

In option ‘a’, no two opposite faces appear together. So, the cube can be formed.

In options other than a, two opposite faces appear together. So, the cubes cannot be formed.

10) Answer: C



11) Answer: B

In the first two dice, we see that Green and Red are the common faces, so definitely, Blue is opposite to Yellow.

12) Answer: A

In the 2nd and 3rd dice, we see that blue and Red are the common faces, so definitely, Violet is opposite to Green.

13) Answer: B

In the first two dice, we see that Y and Z are the common faces, so definitely, B is opposite to X.

14) Answer: C

The faces opposite to each other in the dice are shown below:

6	2	3
1	4	5

15) Answer: A

The faces opposite to each other in the dice are shown below:

A	B	F
D	E	C

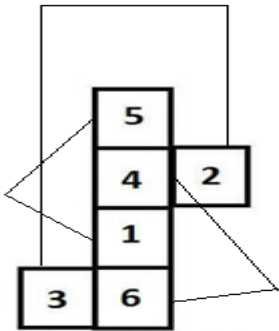
16) Answer: B

As 1 and 3 are common faces in 1st and 3rd dice, so definitely 4 is opposite to 2.

17) Answer: C

As 5 is adjacent to 1, 6, 2 and 3, it is opposite to 4.

18) Answer: B



5, 4, 1 and 6 are to 2. ‘3’ will be opposite to ‘2’, which means 3 will not be adjacent to ‘2’.

19) Answer: B

6 and 2 are the common faces in both the dice, so definitely 2 is opposite to 1.

20) Answer: D

As, ‘1’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

1	2	6
1	5	3

21) Answer: C

As, ‘5’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

5	2	6
5	4	1

22) Answer: C

As, ‘5’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

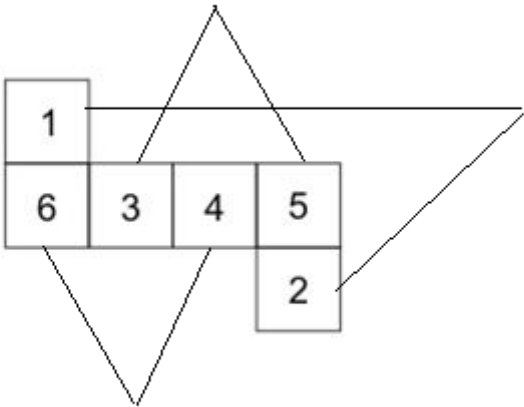
5	3	2
5	6	4

23) Answer: D

As, ‘1’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

1	4	3
1	6	2

24) Answer: D



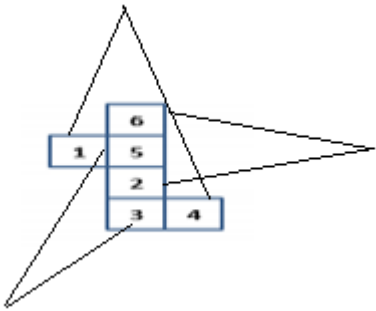
Opposite faces are:

6	3	1
4	5	2

25) Answer: C

As A is adjacent to *, %, # and \$, it is opposite to ‘@’.

26) Answer: D



Opposite faces are:

1	6	5
4	2	3

27) Answer: D

As Red is adjacent to Green, Yellow, Blue and Violet, it is opposite to ‘Pink’.

28) Answer: C

As, ‘Red’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

Red	Green	Blue
Red	Violet	Yellow

29) Answer: B

As, ‘Y’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

Y	C	X
Y	Z	B

30) Answer: B

The faces opposite to each other in the dice is shown below:

1	5	6
1	3	2

31) Answer: A

The faces opposite to each other in the dice is shown below:

3	1	6
3	4	2

As 3 is adjacent to 1, 4, 6 and 2, it is opposite to 5.

32) Answer: B

The faces opposite to each other in the dice is shown below:

#	&	\$
%	*	+

33) Answer: A

As, ‘Y’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

Y	C	X
Y	Z	B

34) Answer: C

The faces opposite to each other in the dice is shown below:

6	2	3
1	4	5

35) Answer: D

The faces opposite to each other in the dice is shown below:

6	2	3
1	4	5

36) Answer: D

The faces opposite to each other in the dice is shown below:

5	4	1
3	6	2

37) Answer: A

The faces opposite to each other in the dice is shown below:

4	6	3
4	5	1

38) Answer: B

The faces opposite to each other in the dice is shown below:

A	B	F
D	E	C

39) Answer: C

As, ‘O’ is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

O	W	E
O	M	N

40) Answer: A

As, 'T' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

T	S	Q
T	L	N

41) Answer: B

As, '@' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

@	\$	!
@	%	*

42) Answer: B

As, 'pk' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

pk	bl	ew
pk	ln	Kt

43) Answer: A

As 7, 2, 5 are adjacent to '8', '8' is opposite to '3'.

44) Answer: A

As, 'O' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

O	W	E
O	M	N

45) Answer: C

As, 'pk' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

pk	bl	ew
pk	ln	kt

46) Answer: A

As, '@' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

@	\$	!
@	%	*

47) Answer: C

As, 'T' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

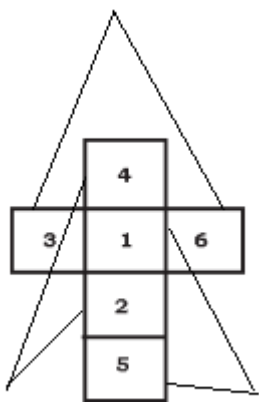
T	S	Q
T	L	N

48) Answer: B

As, '6' is common face in both the dice, keeping it constant and then moving in clockwise direction, we get the faces opposite to each other:

6	5	3
6	1	2

49) Answer: C



Dice 'c' cannot be formed as 3 and 6 are opposite faces.

50) Answer: D

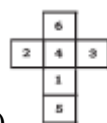
Opposite faces of:



a) $2 + 5 = 7$; $4 + 6 = 10$; $1 + 3 = 4$



b) $1 + 6 = 7$; $5 + 3 = 8$; $4 + 2 = 6$



c) $6 + 1 = 7$; $4 + 5 = 9$; $2 + 3 = 5$



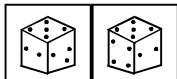
d) $2 + 5 = 7$; $1 + 6 = 7$; $4 + 3 = 7$

So, option d when folded, can create a standard dice.

TYPE-I

1. Two positions of a dice are shown below. When 3 is at the bottom, what number will be at the top?

Question Figures :



- (1) 6 (2) 3
(3) 4 (4) 2

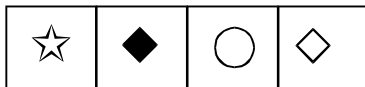
(SSC Combined Graduate Level Prelim Exam. 24.02.2002 (First Sitting))

2. Two positions of a dice are shown below :



When the heart shape is at the top, what will be at the bottom ?

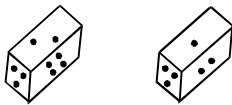
Answer Figures :



- (1) (2) (3) (4)

(SSC Combined Graduate Level Prelim Exam. 24.02.2002 (IInd Sitting))

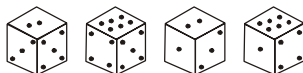
3. Study the two different positions of a dice. When the face containing one dot is at bottom then how many dots would be there on the top face?



- (1) 6 (2) 5
(3) 4 (4) 3

(SSC Combined Graduate Level Prelim Exam. 24.02.2002 (Middle Zone))

4. How many dots are there on the face opposite the face with two dots?

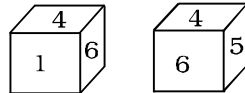


- (1) 1 (2) 5
(3) 4 (4) 6

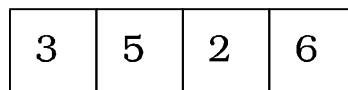
(SSC Combined Matric Level (Pre) Exam. 12.05.2002 (IInd Sitting))

5. Two positions of a dice are shown below. When number 'one' is on the top, what number will be at the bottom?

Question Figures :



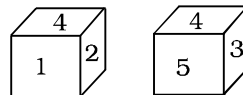
Answer Figures :



- (1) (2) (3) (4)

(SSC Combined Graduate Level Prelim Exam. 11.05.2003 (First Sitting))

6. Two positions of a dice are shown below :

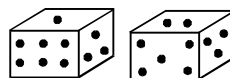


When 3 is at the bottom, which number is at the top ?

- (1) 4 (2) 5
(3) 2 (4) 1

(SSC Combined Graduate Level Prelim Exam. 11.05.2003 (Second Sitting))

7. Two positions of a dice are shown below :

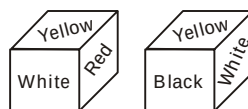


When '2' is at the bottom, what number will be at the top ?

- (1) 3 (2) 5
(3) 1 (4) 6

(SSC Combined Graduate Level Prelim Exam. 08.02.2004 (First Sitting))

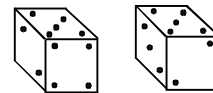
8. From the following two different appearances of die, find out the colour which is opposite to Red :



- (1) Blue (2) Black
(3) White (4) Yellow

(SSC CPO Sub-Inspector Exam. 05.09.2004)

9. Two positions of a dice are shown below :

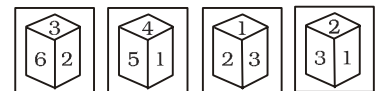


When 4 is at the bottom, what will be at the top?

- (1) 2 (2) 3
(3) 5 (4) 6

(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (1st Sitting) (East Zone))

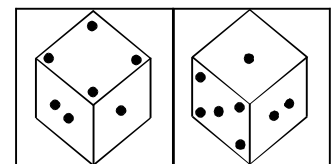
10. Four positions of a dice are given in the question figures. Identify the number at the bottom when top's is 6.



- (1) 5 (2) 1
(3) 3 (4) 4

(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (IInd Sitting) (Middle Zone, Allahabad))

11. Two positions of a dice are shown below. Which number is in front of 5?



- (1) 3 (2) 1
(3) 2 (4) 4

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (1st Sitting))

12. Two positions of a dice are shown below. When 4 is at the top what number will be at the bottom ?

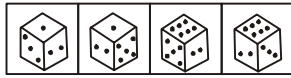


- (1) 5 (2) 1
(3) 2 (4) 6

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (1st Sitting) (Eastern Zone, Guwahati))

CUBES AND DICE

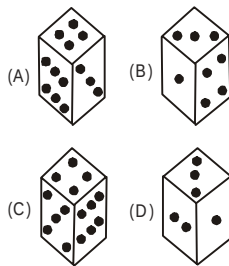
- 13.** How many dots are there on the dice face opposite the face with one dot?



- (1) 2 (2) 3
(3) 4 (4) 6

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (IInd Sitting) (Eastern Zone, Guwahati))

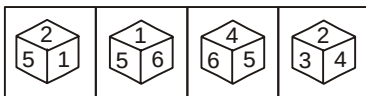
- 14.** How many dots are there on the dice face opposite the face with three dots?



- (1) 2 (2) 4
(3) 5 (4) 6

(SSC Combined Matric Level (Pre) Exam. 05.05.2002 (IInd Sitting) (North Zone Delhi))

- 15.** Observe the faces of the dice and then answer the question.



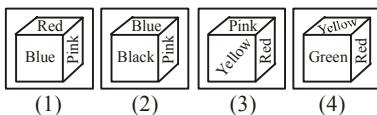
Which number is on the face opposite to 4?

- (1) 1 (2) 2
(3) 3 (4) 5

(SSC Combined Matric Level (Pre) Exam. 12.05.2002 (Ist Sitting))

- 16.** Observe the faces of the dice and answer the question.

Which colour is opposite to Red?

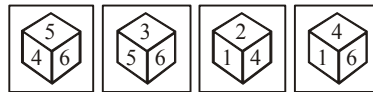


- (1) Yellow (2) Pink
(3) Green (4) Black

(SSC Combined Matric Level (Pre) Exam. 30.07.2006 (Ist Sitting) (East Zone))

- 17.** Four portions of a dice are shown below :

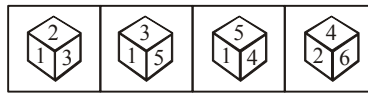
Which number is on the face opposite to 3?



- (1) 1 (2) 2
(3) 4 (4) 5

(SSC Combined Matric Level (Pre) Exam. 30.07.2006 (Ist Sitting) (East Zone))

- 18.** Four positions of a dice are given below.



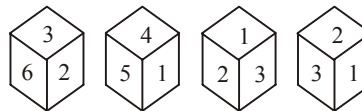
- (1) 1 (2) 2
(3) 4 (4) 5

Identify the number at the bottom when top is 6.

- (1) 3 (2) 1
(3) 4 (4) 5

(SSC Combined Matric Level (Pre) Exam. 30.07.2006 (IInd Sitting) (Central Zone))

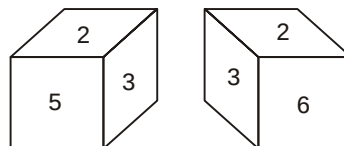
- 19.** Four positions of a dice are given below. Identify the number at the bottom when top's number is 6 :



- (1) 5 (2) 1
(3) 3 (4) 4

(SSC Data Entry Operator Exam. 31.08.2008)

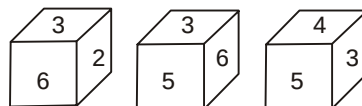
- 20.** From the following two different appearances of die find out the number which is opposite to '5'.



- (1) 2 (2) 3
(3) 4 (4) 6

(SSC Stenographer (Grade 'C' & 'D') Exam. 26.09.2010)

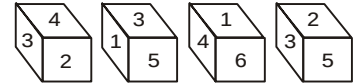
- 21.** Three positions of a dice are given below. Identify the number on the face opposite to 6.



- (1) 1 (2) 4
(3) 5 (4) 6

(SSC Stenographer (Grade 'C' & 'D') Exam. 16.10.2011)

- 22.** Position of Dices is given below :

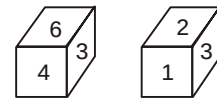


Identify the number when top is 5 what will be at bottom ?

- (1) 6 (2) 3
(3) 4 (4) 2

(SSC Level Data Entry Operator & LDC Exam.04.11.2012 (IInd Sitting))

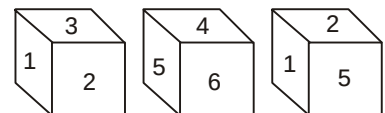
- 23.** Which number appear in the face opposite to the face with number 4 ?



- (1) 1 (2) 2
(3) 3 (4) 5

(SSC Level Data Entry Operator & LDC Exam.04.11.2012 (IInd Sitting))

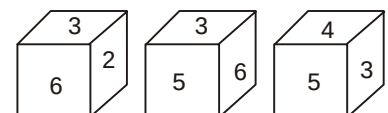
- 24.** Study the following figures and find out the number opposite to 3.



- (1) 6 (2) 4
(3) 5 (4) 2

(SSC Assistant Grade-III Exam.11.11.2012 (IInd Sitting))

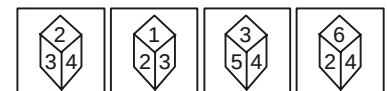
- 25.** Study the following figures and find out the number opposite to 2.



- (1) 1 (2) 4
(3) 5 (4) 6

(SSC Assistant Grade-III Exam.11.11.2012 (IInd Sitting))

- 26.** A dice is thrown four times and its four different positions are given below. Find the number on the face opposite the face showing 2.

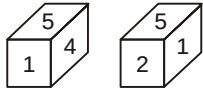


- (1) 4 (2) 5
(3) 6 (4) 3

(SSC (10+2) Level Data Entry Operator & LDC Exam. 04.11.2012, Ist Sitting)

| CUBES AND DICE |

27. From the following two different appearance of a dice, which number lies to the opposite of 2 ?

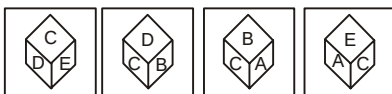


- (1) 1 (2) 4
(3) 6 (4) 5

(SSC (10+2) Level Data Entry Operator & LDC Exam. 04.11.2012, 1st Sitting)

28. Four positions of dice are given below. Which letter will be opposite to D ?

Question Figures :



- (1) B (2) C
(3) D (4) A

(SSC Graduate Level Tier-I Exam. 21.04.2013, 1st Sitting)

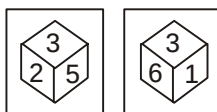
29. Three positions of a dice are given. Find out which number is found opposite the number 2 in the given cube.



- (1) 6 (2) 5
(3) 3 (4) 1

(SSC Graduate Level Tier-I Exam. 21.04.2013, 1st Sitting)

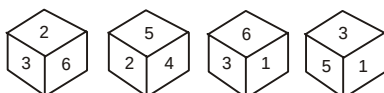
30. Two positions of a dice are given. Which number would be at the top when bottom is 2 ?



- (1) 4 (2) 1
(3) 5 (4) 6

(SSC Graduate Level Tier-I Exam. 21.04.2013, 1st Sitting)

31. Four different positions of dice are as shown below. What number is opposite to face 3 ?

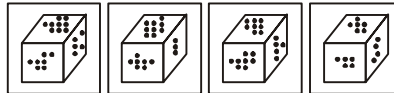


- (1) 4 (2) 3
(3) 2 (4) 6

(SSC CAPFs SI & CISF ASI Exam. 23.06.2013)

32. If the difference between the no. of dots on the opposite faces is 3, find out the figure which is correct ?

Answer Figures :

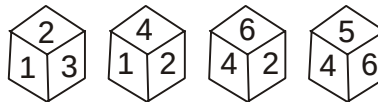


- (1) (2) (3) (4)

(SSC CGL Tier-I Re-Exam-2013, 27.04.2014)

33. Four positions of a dice are given below. Identify the number at the bottom when top is 6.

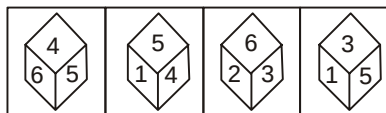
Question Figures



- (1) 1 (2) 3
(3) 4 (4) 5

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam. 22.06.2014)

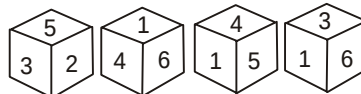
34. Four positions of a dice are given below. Identify the number at the bottom when the number on the top is 2.



- (1) 3 (2) 5
(3) 4 (4) 6

(SSC CAPFs SI, CISF ASI & Delhi Police SI Exam. 22.06.2014)

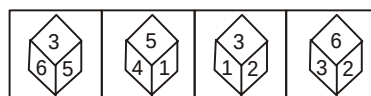
35. Four positions of a dice are given below. Identify the number at the bottom when top is 1.



- (1) 6 (2) 3
(3) 2 (4) 5

(SSC GL Tier-I Re-Exam. (2013) 20.07.2014, 1st Sitting & Bihar SSC 2nd CGL (Pre) Exam. 16.02.2015)

36. Four positions of a dice are given below. Identify the number at the bottom when the number on the top is 2.

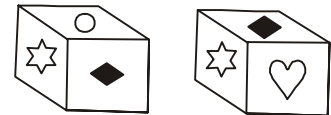


- (1) 6 (2) 3
(3) 4 (4) 5

(SSC GL Tier-I Re-Exam. (2013) 20.07.2014, 1st Sitting)

37. Two positions of a dice are shown below. When the heart shape is at the top what will be at the bottom ?

Question Figures :

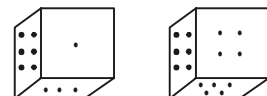


Answer Figures :



(SSC CHSL (10+2) DEO & LDC Exam. 16.11.2014, 1st Sitting TF No. 545 QP 6)

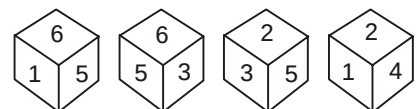
38. Two positions of a dice are shown below. When three is at the top what number will be at the bottom ?



- (1) 2 (2) 1
(3) 4 (4) 5

(SSC CGL Tier-I Exam, 16.08.2015 (1st Sitting) TF No. 3196279)

39. Four positions of a dice are given below :



Find the number on the face opposite to the face showing 4.

- (1) 5 (2) 6
(3) 3 (4) 1

(SSC (10+2) LDC/DEO/PA/SA Exam. 01.11.2015 TF No. 1098066)

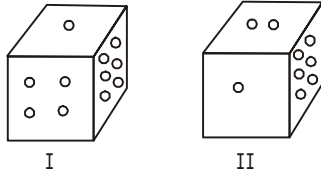
40. Six faces of the dice are A, B, C, D, E and F. A is adjacent to B, B is adjacent to D but not C. E is adjacent to D and F. What is the side opposite to A?

- (1) C (2) D
(3) E (4) F

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 01.11.2015, 1st Sitting)

| CUBES AND DICE |

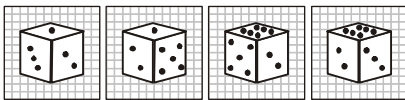
- 41.** In the diagrams below, faces of dice are shown from two different directions. Which number will be opposite to 4?



- (1) 2 (2) 3
(3) 1 (4) 5

(SSC CAPFs (CPO) SI & ASI,
Delhi Police Exam. 05.06.2016)
(1st Sitting)

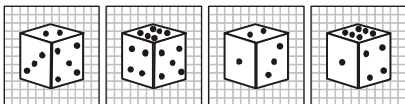
- 42.** Four different views of a cube/dice are given as viewed from different angles. Find out the number of dots on the face opposite to the face with one dot.



- (1) 6 (2) 5
(3) 4 (4) 3

(SSC CGL Tier-I (CBE)
Exam. 08.09.2016) (IIIrd Sitting)

- 43.** How many dots will be there on the face opposite to the face with 2 dots?

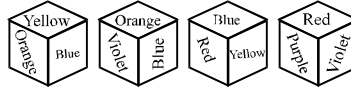


- (1) 1 (2) 5
(3) 4 (4) 6

(SSC CGL Tier-I (CBE)
Exam. 09.09.2016) (IIInd Sitting)

TYPE-II

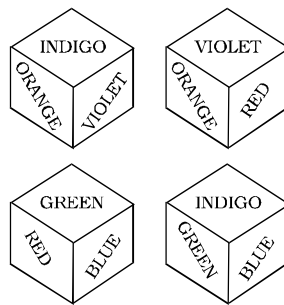
- 1.** Which colour is opposite to yellow?



- (1) Violet (2) Red
(3) Purple (4) Blue

(SSC Combined Graduate Level
Prelim Exam. 04.07.1999 (First
Sitting))

- 2.** Which colour is opposite to Green?



- (1) Indigo (2) Blue
(3) Violet (4) Orange

(SSC Combined Graduate Level
Prelim Exam. 04.07.1999
(Second Sitting))

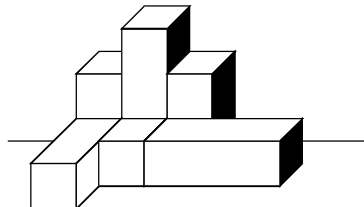
Direction (3) : Study the two different positions of a cube given below with dots from 1 to 6 marked on its faces. When 3 is at the bottom, what will be at the top?

(SSC CPO Sub-Inspector
Exam.12.01.2003)



- (1) 2 (2) 3
(3) 4 (4) 6

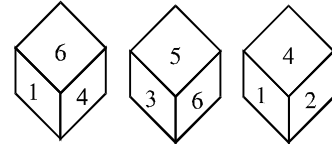
- 4.** Please count the number of blocks in the given figure and indicate :



- (1) 9 (2) 5
(3) 6 (4) 8

(SSC CPO Sub-Inspector
Exam. 07.09.2003)

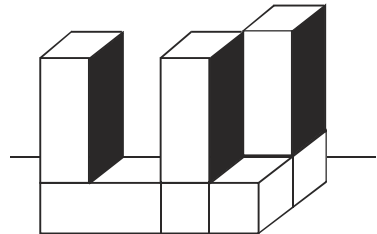
- 5.** Three positions of a cube are given. Based on them find out which number is found opposite of number 2 in a given cube?



- (1) 2 (2) 5
(3) 1 (4) 6

(SSC Combined Graduate Level
Prelim Exam. 08.02.2004
(Second Sitting))

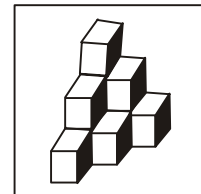
- 6.** Please count the number of blocks in the given figure and indicate.



- (1) 6 (2) 8
(3) 9 (4) 7

(SSC CPO Sub-Inspector
Exam. 05.09.2004)

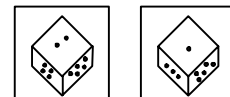
- 7.** Count the number of cubes, in the given figure.



- (1) 8 (2) 10
(3) 12 (4) 14

(SSC Statistical Investigators
Grade-IV Exam. 31.07.2005)

- 8.** Study the two different positions of a cube given below with dots from 1 to 6 marked on its faces. Find out how many dots are contained on the face opposite to that containing 4 dots.

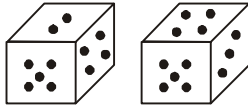


- (1) 1 (2) 2
(3) 3 (4) 5

(SSC Combined Graduate Level
Prelim Exam. 13.11.2005
(First Sitting))

| CUBES AND DICE |

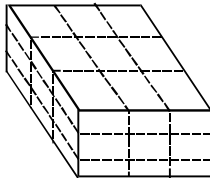
9. Study the two different positions of a cube given below with dots from 1 to 6 marked on its faces. Find out how many dots are contained on the face opposite to that containing 3 dots.



- (1) 6 (2) 2
(3) 1 (4) 5

(SSC Combined Graduate Level Prelim Exam. 13.11.2005 (Second Sitting))

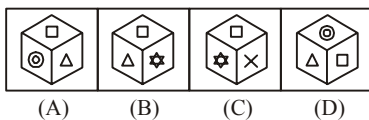
Direction (10) : The cube given below has been painted by three different colours. The opposite surfaces have been painted by the same colour. Next, the cube has been cut into 27 equal parts. Answer the following questions on the basis of information so provided.



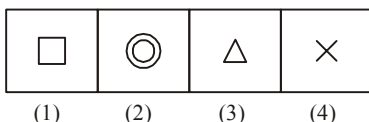
(SSC Statistical Investigators Grade-IV Exam. 13.08.2006)

10. How many such small cubes will be there whose only one surface is painted ?
(1) 4 (2) 6
(3) 8 (4) 12
11. Given below are four views of a cube. Each face is marked with a certain symbol. In figure B which symbol will appear on the face opposite to the face having symbol ?

Question Figures :



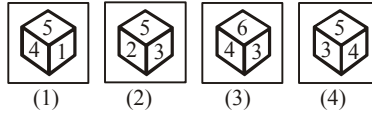
Answer Figures :



- (1) (2) (3) (4)

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (First Sitting))

12. Four views of one cube are given below. Each of the six faces of cube is numbered. In figure 3 which number will be opposite to the face with number 3 ?

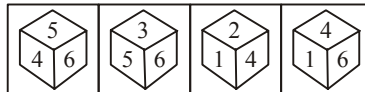


- (1) 1 (2) 2
(3) 5 (4) 6

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (Second Sitting))

13. Study the four different positions of a cube given below with numbers from 1 to 6 marked on its faces. Find out which number is contained on the face opposite to that containing 3.

Question Figures :

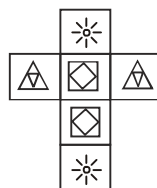


- (1) 5 (2) 4
(3) 2 (4) 6

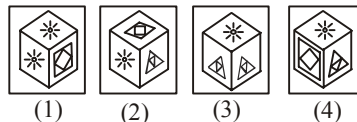
(SSC CPO Sub-Inspector Exam. 16.12.2007)

14. Choose from the four answer figures the figure that will be formed when question figure is folded into a box.

Question Figure :



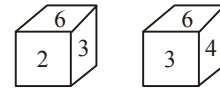
Answer Figures :



- (1) (2) (3) (4)

(SSC Combined Graduate Level Prelim Exam. 27.07.2008 (First Sitting))

15. Two positions of a cube are given. Based on them find out which number is found opposite number 4 in a given cube ?

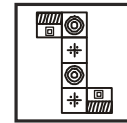


- (1) 1 (2) 2
(3) 3 (4) 4

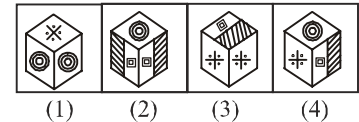
(SSC Combined Graduate Level Prelim Exam. 27.07.2008 (First Sitting))

16. Choose from the four answer figures, the figure that will be formed when question figure is folded into a box.

Question Figure :



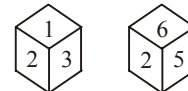
Answer Figures :



- (1) (2) (3) (4)

(SSC Combined Graduate Level Prelim Exam. 27.07.2008 (Second- Sitting))

17. Where is the invisible number in the two positions of the same cube ?

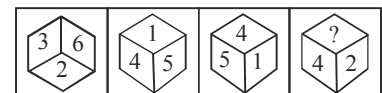


- (1) Opposite of 2
(2) Opposite of 3
(3) Opposite of 4
(4) Opposite of 6

(SSC Combined Graduate Level Prelim Exam. 27.07.2008 (Second- Sitting))

18. The following diagram depicts various views of a cube. Each faces has some number, where as in cube 4, one face is blank. From the answer choices select the number that should come in the blank space.

Question Figures :

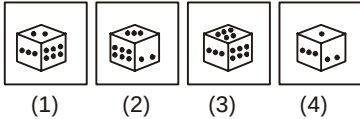


- (1) 1 (2) 5
(3) 6 (4) 3

(SSC CPO Sub-Inspector Exam. 09.11.2008)

CUBES AND DICE

19. The following diagram shows the different view of same cube. Find out how many spots are on face directly opposite to the face of the cube having 6 spots :



- (1) 1 (2) 3
(3) 5 (4) 2

(SSC CPO Sub-Inspector Exam. 06.09.2009)

20. Little wooden cubes each with a side of one inch are put together to form a solid cube with a side of three inches. This big cube is then painted red all over on the outside. When the big cube is broken up into the original little ones, how many cubes will have paint on two sides ?

- (1) 4 (2) 8
(3) 12 (4) 0

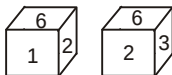
(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (First Sitting))

21. A cube has the following figures drawn on its five faces. The top surface is blank. The ellipse is between the cross and the triangle. The square is on the right of the triangle. The ellipse and the square are opposite to each other. Which face is the circle on?

- (1) On the top
(2) Opposite to ellipse
(3) Opposite to triangle
(4) At the bottom

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (Second Sitting))

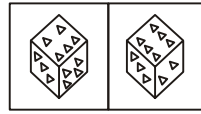
22. Two positions of a cube are given. Based on them find out which number is found opposite number 1 in the given cube.



- (1) 1 (2) 2
(3) 3 (4) 4

(SSC CISF ASI Exam. 29.08.2010 (Paper-I))

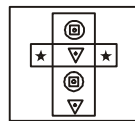
23. Two positions of a cubical block are given below, each face having a number of small triangles. In another position of the cube, if there is one triangle at the bottom, how many triangles will be there on the top face ?



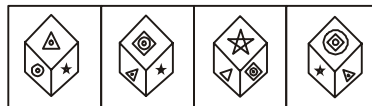
- (1) 4 (2) 3
(3) 2 (4) 5

(SSC CPO Sub-Inspector Exam. 12.12.2010 (Paper-I))

24. Choose from the four answer figures, the figure that will be formed when the question figure is fold into a box.



Answer figures

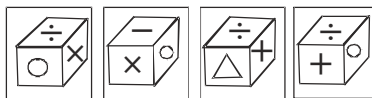


- (1) (2) (3) (4)

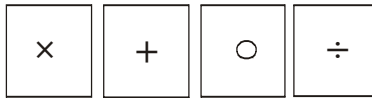
(SSC CPO Sub-Inspector Exam. 12.12.2010 (Paper-I))

25. Four positions of a cube are shown below. Which symbol is opposite the face having "Δ"?

Questions Figures :



Answer Figures :



- (1) (2) (3) (4)

(SSC Combined Matric Level (PRE) Exam. 24.10.1999 (1st Sitting))

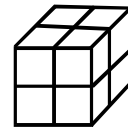
26. Four positions of a cube are shown below. Which symbol is opposite the having '+' ?



- (1) Δ (2) ×
(3) - (4) ÷

(SSC Combined Matric Level (PRE) Exam. 24.10.1999 (IInd Sitting))

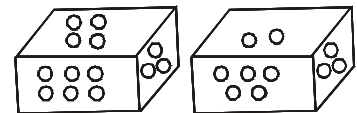
27. In the figure you can see the picture of a group of blocks arranged in a particular way. All the blocks are of same size. You can see only some of the blocks in the group. Others are hidden by the blocks in front of them. Try to count the blocks in the group including those which you cannot see.



- (1) 4 (2) 8
(3) 9 (4) 6

(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (1st Sitting) (East Zone))

28. Two positions of a block are given below :

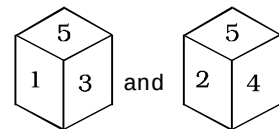


When two is at the bottom what number will be at the top?

- (1) 6 (2) 5
(3) 4 (4) 1

(SSC Combined Matric Level (PRE) Exam. 13.05.2001 (IInd Sitting))

29. Two positions of a block are shown below



When 5 is on top, what is at the bottom?

- (1) 6 (2) 4
(3) 3 (4) 2

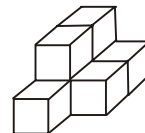
(SSC Combined Matric Level (PRE) Exam. 27.05.2001 (IInd Sitting) (East Zone))

30. A cube is painted red on all the six sides. It is then cut into 27 equal cubes. How many cubes are coloured on one face only ?

- (1) 0 (2) 6
(3) 8 (4) 18

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (1st Sitting) (Eastern Zone, Guwahati))

31. The figure below is a drawing of a pile of blocks. When taken apart, how many blocks would be there ?

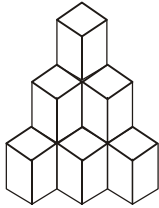


- (1) 3 (2) 5
(3) 7 (4) 13

(SSC Combined Matric Level (PRE) Exam. 05.05.2002 (1st Sitting) (Eastern Zone, Guwahati))

CUBES AND DICE

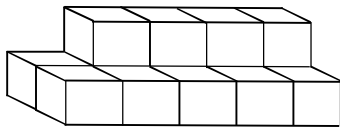
32. How many cubes are there in the given (figure) ?



- (1) 6 (2) 10
(3) 12 (4) 8

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (IInd Sitting)
(Eastern Zone, Guwahati)

33.

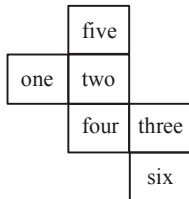


The figure above is a drawing of pile of Blocks. When taken apart, how many blocks would be there?

- (1) 15 (2) 14
(3) 10 (4) 5

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

34.



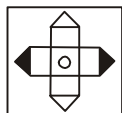
If a cube is made using above arrangement which word will be in opposite face of the word one?

- (1) Three (2) Six
(3) Four (4) Five

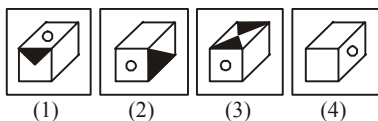
(SSC Combined Matric Level (Pre)
Exam. 30.07.2006 (Ist Sitting)
(East Zone)

35. Which of the four cubes given below can be created by folding the given design?

Question Figure :



Answer Figures :



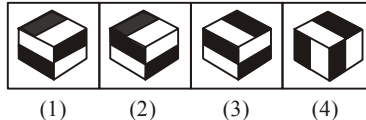
(1) (2) (3) (4)
(SSC Combined Matric Level (Pre)
Exam. 30.07.2006 (Ist Sitting)
(East Zone)

36. In this test the problem figure is of a wooden block. Think of the position which you will get when you place the block with its front side at the bottom.

Question Figure :

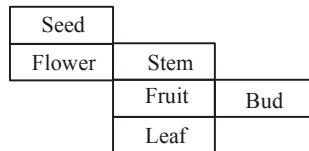


Answer Figures :



(1) (2) (3) (4)
(SSC Combined Matric Level (Pre) Exam.
30.07.2006 (IInd Sitting) (Central Zone)

37. In a cube using given arrangement, which part be opposite Fruit?

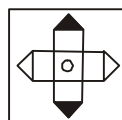


- (1) Stem (2) Seed
(3) Flower (4) Bud

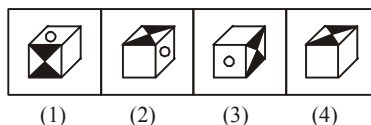
(SSC Combined Matric Level (Pre) Exam.
30.07.2006 (IInd Sitting) (Central Zone)

38. The question figure is folded to form a box. Choose from the answer figures the box that is similar to the box formed.

Question Figure :

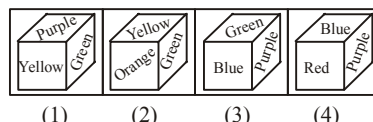


Answer Figures :



(1) (2) (3) (4)
(SSC Combined Matric Level (Pre) Exam.
30.07.2006 (IInd Sitting) (Central Zone)

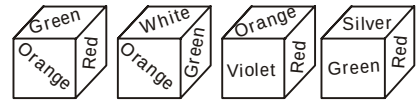
39. Which colour is opposite to Purple?



- (1) Blue (2) Orange
(3) Red (4) Green

(SSC Combined Matric Level (Pre) Exam.
30.07.2006 (IInd Sitting) (Central Zone)

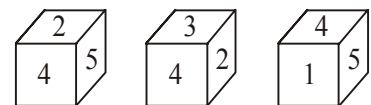
40. If a cube is made using the given arrangements, which colour will be opposite the colour Green ?



- (1) Orange (2) Red
(3) Silver (4) Violet

(SSC Combined Matric Level (Pre)
Exam. 30.03.2008 (Ist Sitting)

41. Which number will appear at the bottom face in last cube ?



- (1) 3 (2) 4
(3) 6 (4) 1

(SSC Combined Matric Level (Pre)
Exam. 30.03.2008 (Ist Sitting)

42. In a solid cube which is made up of 27 small cubes, two opposite sides are painted red, two painted yellow and the other two with white. How many cubes have two colours ?

- (1) 8 (2) 12
(3) 16 (4) 24

(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 27.11.2010)

43. Wooden little cubes each with an edge of one inch are put together to form a solid cube with an edge of three inches. This big cube is then painted red all over the outside. When the big cube is broken-up into the original little ones, how many cubes will be without paint?

- (1) 0 (2) 1
(3) 3 (4) 4

(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 28.11.2010 (Ist sitting)

44. A solid cube is made using 64 small cubes. In how many small cubes two sides are seen ?

- (1) 24 (2) 32
(3) 40 (4) 42

(SSC Higher Secondary Level
Data Entry Operator & LDC Exam.
28.11.2010 (Ist sitting)

| CUBES AND DICE |

- 45.** In a solid cube made up of 27 small cubes, two opposite sides are painted red, two opposite sides yellow and two other sides white. How many small cubes have the colours yellow and white alone in them ?

(1) 4 (2) 8
(3) 12 (4) 16

(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 28.11.2010 (IInd sitting))

- 46.** A solid cube of 4 inches has been painted Red, Green, and Black on pair of opposite faces. It has been cut into one inch cubes. How many cubes have only one face painted that too only red ?

(1) 4 (2) 8
(3) 16 (4) 24

(SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(Ist Sitting (North Zone))

- 47.** A solid cube of 4 inches has been painted red, green and black on pairs of opposite faces. It has been cut into one inch cubes. How many cubes have only three faces painted ?

(1) 4 (2) 8
(3) 12 (4) 16

(SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(IInd Sitting (North Zone))

- 48.** A solid cube of 4 inches has been painted Red, Green and Black on pair of opposite faces. It has then been cut into one inch cubes. How many cubes have only four faces painted?

(1) 0 (2) 4
(3) 8 (4) 16

(SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(Ist Sitting (East Zone))

- 49.** A solid cube of 4 inches has been painted red, green and black on pairs of opposite faces. It has been cut into one inch cubes. How many cubes have all three colours?

(1) 0 (2) 4
(3) 8 (4) 16

(SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(IInd Sitting (East Zone))

- 50.** A solid cube of 4 inches has been painted red, green and black on pair of opposite faces. It has been cut into one inch cubes. How many cubes have only Red and Green faces?

(1) 4 (2) 8
(3) 16 (4) 24

(SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(Ist Sitting (Delhi Zone))

- 51.** A solid cube of 4 inches has been painted red, green and black on pair of opposite faces. It has been cut into one inch cubes. How many cubes have only two faces painted ?

(1) 8 (2) 16
(3) 24 (4) 32

(SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(IInd Sitting (Delhi Zone))

- 52.** A solid cube of 4 inches has been painted red, green and black on pair of opposite faces. It has been cut into one inch cubes. How many cubes have no face painted ?

(1) 0 (2) 4
(3) 8 (4) 16

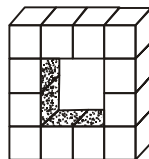
(SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(Ist Sitting (East Zone))

- 53.** A solid cube of 4 inches has been painted red, green and black on pair of opposite faces. It has been cut into one inch cubes. How many cubes have only one face painted ?

(1) 4 (2) 8
(3) 16 (4) 24

(SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(IInd Sitting (East Zone))

- 54.** How many cubes are there in this diagram ?



(1) 10 (2) 8
(3) 16 (4) 12

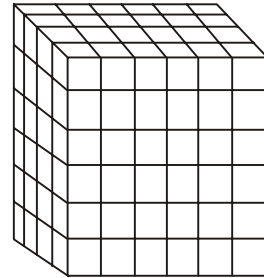
(SSC Level Data Entry Operator &
LDC Exam. 21.10.2012 (Ist Sitting))

- 55.** A solid red coloured cube is painted yellow on all sides. The cube is cut into 125 equal cubes. How many sides will have 3 sides yellow ?

(1) 10 (2) 4
(3) 8 (4) 12

(SSC Level Data Entry Operator &
LDC Exam. 21.10.2012 (IInd Sitting))

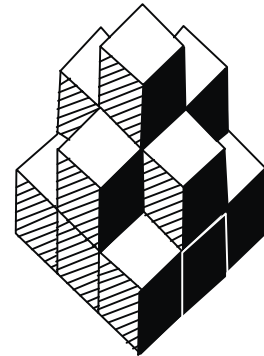
- 56.** How many cubes are there in this figure ?



(1) 69 (2) 180
(3) 144 (4) 84

(SSC Level Data Entry Operator &
LDC Exam. 21.10.2012 (IInd Sitting))

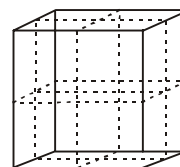
- 57.** How many cubes are unseen in the figure ?



(1) 5 (2) 6
(3) 10 (4) 15

(SSC Assistant Grade-III
Exam. 11.11.2012 (IInd Sitting))

- 58.** A cube which is painted red on the outer surface is of 2 inches height, 2 inches wide and 2 inches across. If it is cut into one-inch cubes as shown by dotted lines, indicate the number of cubes which are red on two sides only ?



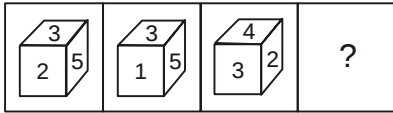
(1) 4 (2) 6
(3) 8 (4) 0

(SSC Graduate Level Tier-I
Exam. 21.04.2013, Ist Sitting)

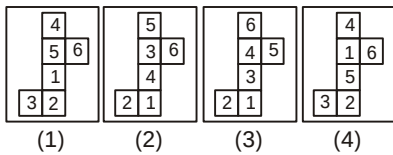
| CUBES AND DICE |

- 59.** Three views of the same cube are given. All the faces of the cube are numbered from 1 to 6. Select one figure which will result when the cube is unfolded.

Question Figures :



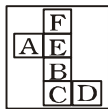
Answer Figures :



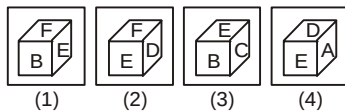
(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

- 60.** Which of the following cubes can be created by folding the given figure?

Question Figure :

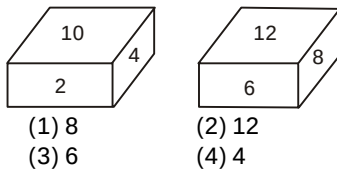


Answer Figures :



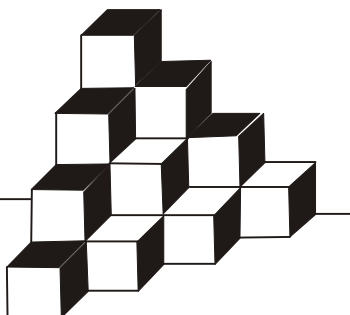
(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

- 61.** From the given blocks when 10 is at the bottom, which number will be at the top ?



(SSC Graduate Level Tier-I
Exam. 19.05.2013, 1st Sitting)

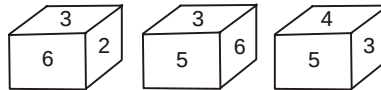
- 62.** How many cubes are there in the group ?



- (1) 20 (2) 10
(3) 16 (4) 18

(SSC Graduate Level Tier-I
Exam. 19.05.2013, 1st Sitting)

- 63.** Study the 3 different positions of a cube given below and answer what number comes opposite to 2 ?

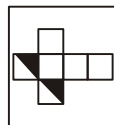


- (1) 4 (2) 5
(3) 3 (4) 1

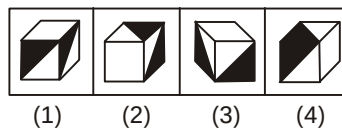
(SSC CAPFs SI & CISF ASI
Exam. 23.06.2013)

- 64.** Choose the cube which will be formed on folding the given question figure.

Question Figure :

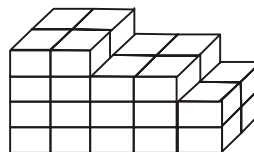


Answer Figures :



(SSC GL Tier-I Exam. 26.10.2014)

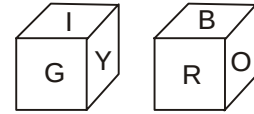
- 65.** Few cubes are arranged as shown in the figure. How many cubes are unseen ?



- (1) 8 (2) 10
(3) 12 (4) 14

(SSC CHSL (10+2) DEO & LDC
Exam. 02.11.2014, Patna Region :
1st Sitting)

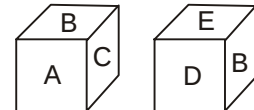
- 66.** The sides of a cube show the colours of rainbow. Two positions of the cube are shown below. Which of the colours of rainbow is left out ?



- (1) Yellow (2) Green
(3) Violet (4) Indigo

(SSC CHSL (10+2) DEO & LDC
Exam. 02.11.2014, 1st Sitting)

- 67.** Based on the diagrams which letter is opposite to A ?

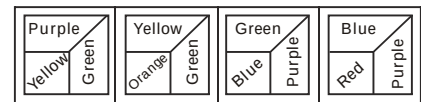


- (1) B (2) C
(3) D (4) E

(SSC CHSL (10+2) DEO & LDC
Exam. 02.11.2014, 1st Sitting)

- 68.** Which colour is opposite to purple ?

Question Figures :

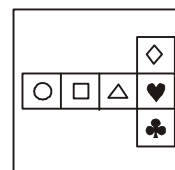


- (1) Blue (2) Orange
(3) Red (4) Green

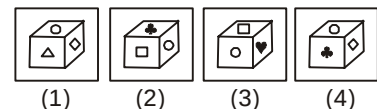
(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Patna Region :
1st Sitting)

- 69.** Choose the cube that will be formed by folding the sheet of paper shown in the problem figure.

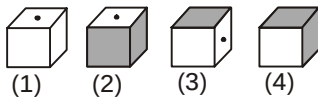
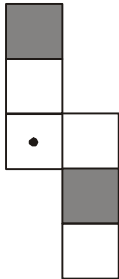
Question Figure :



Answer Figures :



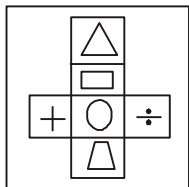
70. The figure given on the left hand side is folded to form a box. Choose from the alternatives (1), (2), (3) and (4) the boxes that is similar to the box formed.



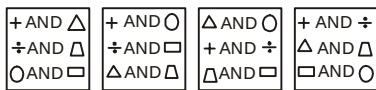
- (1) (2) and (3) only
(2) (1), (3) and (4) only
(3) (2) and (4) only
(4) (1) and (4) only

71. If a paper is folded as shown in figure to form a cube, then the pairs of opposite faces are :

Question Figure :

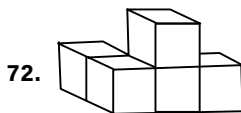


Answer Figures :



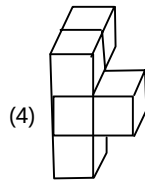
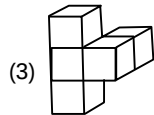
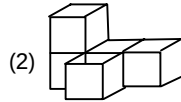
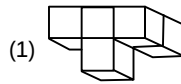
- (1) (2) (3) (4)

(SSC CGL Tier-I Exam, 16.08.2015
(IInd Sitting) TF No. 2176783)



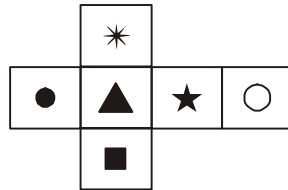
72.

The solid so formed by joining unit cubes is rotated to obtain different positions, which of these cannot be the shape after it has turned?



(SSC CHSL (10+2) LDC, DEO
& PA/SA Exam, 15.11.2015
(IInd Sitting) TF No. 7203752)

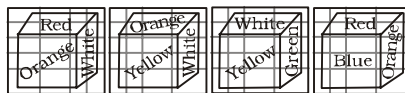
73. The following figure is folded to form a block. Which symbol will appear on the opposite of ?



- (1) (2)
(3) (4)

(SSC (10+2) Stenographer Grade
'C' & 'D' Exam. 31.01.2016
TF No. 3513283)

74. Four positions of a cube are shown below. Which colour is opposite to white colour in the given cubes?

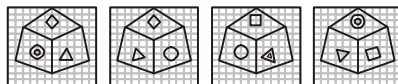


- (1) Orange (2) Blue
(3) Red (4) Yellow

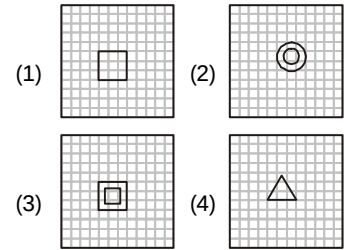
(SSC CGL Tier-I (CBE) Exam.10.09.2016)

75. Which symbol will appear on the face opposite to the face of a circle O in the cube given below?

Question Figures :

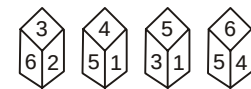


Answer Figures :



(SSC CGL Tier-I (CBE)
Exam. 09.09.2016) (Ist Sitting)

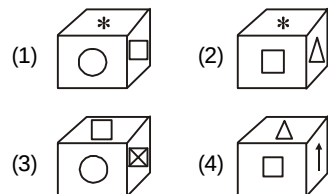
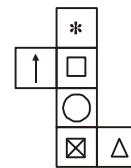
76. Four portions of a cube are shown below. Identify the numbers at the bottom when top is 6 ?



- (1) 6 (2) 2
(3) 1 (4) 5

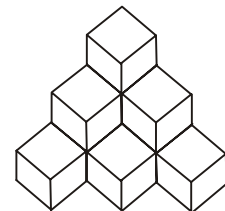
(SSC CAPFs (CPO) SI & ASI,
Delhi Police Exam. 20.03.2016)
(IInd Sitting)

77. Which of the following cubes can be created by folding the given figure ?



(SSC CPO SI & ASI, Online
Exam. 06.06.2016) (IInd Sitting)

78. How many cubes are there in the diagram ?

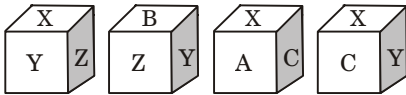


- (1) 10 (2) 12
(3) 8 (4) 6

(SSC CGL Tier-I (CBE)
Exam. 30.08.2016) (Ist Sitting)

CUBES AND DICE

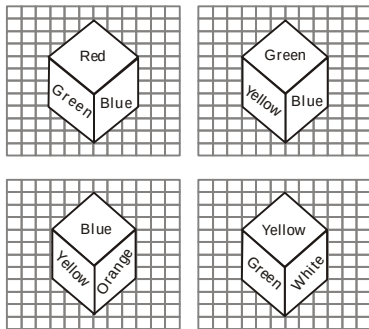
- 79.** Of a cubical container, all the six faces have different markings indicating the position of the equipment packed inside. Which symbol is opposite the face having C?



- (1) Z (2) B
(3) X (4) Y

(SSC CGL Tier-I (CBE)
Exam. 03.09.2016) (IInd Sitting)

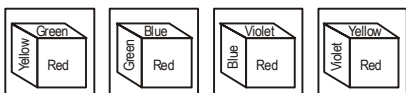
- 80.** Four positions of a cube are shown in the diagram. Which colour is opposite to Green colour in the given cubes?



- (1) Blue (2) Yellow
(3) Orange (4) White

(SSC CGL Tier-I (CBE)
Exam. 06.09.2016) (Ist Sitting)

- 81.** In the given cube, which colour is opposite to Blue ?



- (1) Green (2) Red
(3) Violet (4) Yellow

(SSC CGL Tier-I (CBE)
Exam. 28.08.2016) (Ist Sitting)

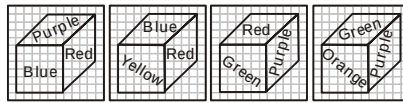
- 82.** From the positions of a cube shown below, which letter will be on the face opposite to face with 'A'?



- (1) F (2) D
(3) C (4) B

(SSC CGL Tier-I (CBE)
Exam. 10.09.2016) (IIIrd Sitting)

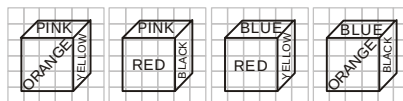
- 83.** In the given cubes, which colour is opposite to purple?



- (1) Violet (2) Red
(3) Yellow (4) Blue

(SSC CGL Tier-I (CBE)
Exam. 09.09.2016) (IInd Sitting)

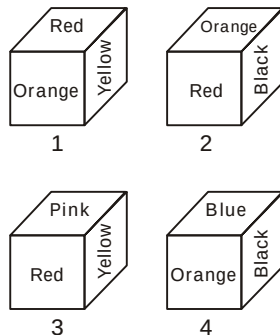
- 84.** Six faces of the cube are painted with different colours as shown below. In figure 3, which colour is opposite to RED?



- (1) Black (2) Yellow
(3) Orange (4) Pink

(SSC CGL Tier-I (CBE)
Exam. 10.09.2016) (IInd Sitting)

- 85.** A block is painted yellow, red, black, orange, pink and blue on its six sides as shown in the following four figures. In figure 3, which colour will be opposite to red?



- (1) Blue (2) Orange
(3) Yellow (4) Pink

(SSC CGL Tier-I (CBE)
Exam. 27.10.2016) (Ist Sitting)

25. (3)	26. (2)	27. (2)	28. (4)
29. (1)	30. (4)	31. (1)	32. (4)
33. (1)	34. (2)	35. (3)	36. (4)
37. (3)	38. (4)	39. (1)	40. (3)
41. (1)	42. (1)	43. (4)	

TYPE-II

1. (1)	2. (4)	3. (4)	4. (3)
5. (4)	6. (4)	7. (2)	8. (3)
9. (2)	10. (2)	11. (2)	12. (1)
13. (2)	14. (4)	15. (2)	16. (4)
17. (1)	18. (4)	19. (1)	20. (2)
21. (4)	22. (3)	23. (2)	24. (2)
25. (3)	26. (2)	27. (2)	28. (3)
29. (1)	30. (2)	31. (3)	32. (2)
33. (2)	34. (1)	35. (4)	36. (3)
37. (2)	38. (4)	39. (2)	40. (4)
41. (3)	42. (2)	43. (2)	44. (1)
45. (1)	46. (2)	47. (2)	48. (1)
49. (3)	50. (3)	51. (3)	52. (3)
53. (4)	54. (4)	55. (3)	56. (3)
57. (1)	58. (4)	59. (4)	60. (2)
61. (2)	62. (1)	63. (2)	64. (3)
65. (2)	66. (3)	67. (3)	68. (2)
69. (2)	70. (2)	71. (3)	72. (3)
73. (3)	74. (2)	75. (2)	76. (3)
77. (2)	78. (1)	79. (1)	80. (3)
81. (4)	82. (2)	83. (3)	84. (3)
85. (1)			

ANSWERS

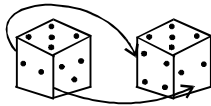
TYPE-I

1. (3)	2. (4)	3. (3)	4. (4)
5. (2)	6. (3)	7. (3)	8. (2)
9. (2)	10. (2)	11. (4)	12. (3)
13. (4)	14. (3)	15. (1)	16. (4)
17. (3)	18. (2)	19. (2)	20. (4)
21. (2)	22. (3)	23. (1)	24. (3)

EXPLANATIONS

TYPE-I

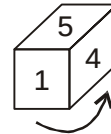
1. (3) From the two views of the dice it is clear that number 4 will be at the top when 3 is at the bottom.



2. (4) When the heart shape is at the top, the rhombus would be on the bottom.
3. (3) When both 2 and 3 dots are at the same side, 1 dot should be opposite to 4 dot.
4. (4) There are 2, 3, 4 and 6 dots respectively on the adjacent faces of the face having 5 dots. Therefore, the face with one dot will be opposite the face having 5 dots.
There are 1, 3, and 5 dots respectively on the adjacent faces of the face having two dots. Therefore, either 4 or 6 dots will be on the face opposite of the face having two dots. Now, 4 and 6 dots cannot be on the opposite faces. Clearly, 6 dots are there on the face opposite to the face with two dots.
5. (2) When both 6 and 4 are in the same position of cube, 5 should be opposite to 1.
6. (3) 1 lies opposite 5.
2 lies opposite 3.
4 lies opposite 6.
7. (3) When '2' is at the bottom, number '1' will be on the top.
8. (2) From the two views of the dice, it is clear that Black is opposite to Red.
9. (2) From the two views of the dice, it is clear that when 4 is at the bottom, 3 will be on the top.
10. (2) From the given views of dice it is clear that the numbers 2, 3, 4, and 5 cannot be on the face opposite 1. Hence, 6 must be on face opposite 1. Therefore, when 6 is on the top face 1 must be on the bottom face.
11. (4) The number 4 will lie opposite 5.

12. (3) The numbers 2, 4 and 5 cannot be on the opposite face of 6. From the two views of dice, it is clear that 2 is at bottom when 4 is at top.
13. (4) There are respectively one, three, five and six dots on faces adjacent to the face having two dots. So, four dots lies on the face opposite to the two dots. Therefore, six dots are on the face opposite to one dot.
14. (3) Dots, one two four and six cannot be on the faces opposite to face containing three dots.
Therefore, five dots are on the face opposite to face with three dots.
15. (1) The numbers 1, 2, 4 and 6 are on the adjacent faces of number 5.
Therefore, number 3 will lie opposite 5.
The numbers 2, 5 and 6 are on the adjacent faces of number 1.
Therefore, 4 lies opposite 1.
16. (4) Blue, Pink, Yellow and Green colours are on the faces adjacent to Red. Therefore, Black colour is opposite to Red.
17. (3) The numbers 1, 2, 5 and 6 lie on the adjacent faces of the number 4. Therefore, the number 3 lies opposite to 4.
18. (2) The numbers 2, 3, 4 and 5 cannot be on the opposite faces of the number 1. Therefore 1 lies opposite to 6.
19. (2) The numbers 2, 3, 4 and 5 cannot be on the face opposite to 1. Therefore, 6 lies opposite 1.
20. (4) From the two different views of the dice it is clear that '6' lies opposite to '5'.
21. (2) The numbers 2, 4, 5 and 6 cannot be on the face opposite to 3.
The numbers 1, 3, 4 and 6 cannot be on the face opposite to 5.
Therefore, 2 lies opposite 5.
Clearly, 4 lies opposite 6.
22. (3) The numbers 1, 2, 4 and 5 lie on the faces adjacent to 3.
Therefore, the number 6 lies opposite 3.
The numbers 3, 4, 5 and 6 lie on the faces adjacent to 1.
Therefore, the number 2 lies opposite 1.
Now, the number 4 lies opposite 5.

23. (1) From the two views of dice, it is clear that number '1' lies opposite to number '4'.
24. (3) The numbers 1, 2, 4 and 6 are on adjacent faces of the number 5. Therefore, the number 3 lies opposite to 5.
25. (3) The numbers 2, 4, 5 and 6 are on adjacent faces of the number 3. Therefore, number 1 lies opposite to 3.
The numbers 2, 3 and 5 are on adjacent faces of the number 6. Therefore, the number 4 lies opposite to 6.
Now, the number 5 lies opposite to 2.
26. (2) The numbers 1, 3, 4 and 6 lie on the faces adjacent to the number 2. Therefore, the number 5 lies on the face opposite to the number 2.
27. (2) Clearly, the number 4 lies to the opposite of 2.



28. (4) The letters A, B, D and E are on the adjacent faces of face C. Therefore, some other letter lies opposite 'C'. The letters B, C and E are on the adjacent faces of face D. Therefore, A lies opposite D.
29. (1) The numbers 1, 3, 4 and 5 are on the adjacent faces of number 6. Therefore, 2 lies opposite 6.
30. (4) The numbers 1, 2, 5 and 6 are on the adjacent faces of number 3. Therefore, the number 4 lies opposite 3.
The numbers 3, 4 and 6 cannot be on the faces opposite to 1. Therefore, 5 lies opposite 1.
Now, 2 lies opposite 6.
31. (1) The numbers 1, 2, 5 and 6 are on the adjacent faces of the number 3. So, the number 4 lies opposite 3.

32. (4) Answer Figure (4) shows the number of dots on its faces like a standard dice. So, in this dice the difference between the numbers of dots on opposite faces may be 3.

33. (1) The numbers 1, 2, 5 and 6 are on the faces adjacent to the number 4. So, the number 3 lies opposite 4.

The numbers 1, 3, 4 and 6 lie on the faces adjacent to the number 2. So, the number 5 lies opposite 2.

Now, the number 1 lies opposite 6.

34. (2) The numbers 1, 3, 4 and 6 are on faces adjacent to number 5. Therefore, 5 lies opposite 2.

35. (3) The numbers 3, 4, 5 and 6 are on the faces adjacent to the number 1. So, 2 lies opposite 1.

36. (4) The numbers 1, 2, 5 and 6 are on the faces adjacent to the number 3. Therefore, 4 lies opposite 3.

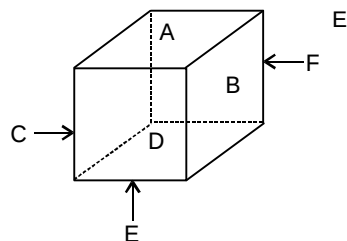
The numbers 2, 3, 4 and 5 are on the faces adjacent to the number 1. Therefore, 6 lies opposite 1. Now, the number 5 lies opposite 2.

37. (3) From the given positions of a dice, it is clear that circle would be at the bottom, when the heart shape is at the top.

38. (4) Six dots are on the face adjacent to one dot, three dots, four dots and five dots. So, two dots lie opposite six dots. If we rotate the first dice downward two times, we get the second dice. So, five dots lie opposite three dots.

39. (1) The numbers 1, 2, 3 and 6 lie on the faces adjacent to the number 5. Therefore, the number 5 lies opposite 4.

40. (3)



is opposite to A.

41. (1) It is clear from the two views of the same dice that from first figure to second figure, the dice has been rotated downward. Therefore, the number 2 lies opposite the number 4.

42. (1) One dot, three dots, five dots and six dots is/are on the faces adjacent to two dots. Therefore, four dots are on face opposite to two dots.

One dot, two dots and six dots is/are on the faces adjacent to three dots. Therefore, five dots are on the face opposite to three dots.

Thus, six dots lie opposite one dot.

43. (4) Two, three, four and six dots are on the faces adjacent to one dot. Therefore, five dots are on face opposite one dot. One dot, three dots and five dots are on the faces adjacent to two dots. From the second view of the cube it is clear that four dots are on the face adjacent to two dots.

Therefore, six dots are on the face opposite two dots.

TYPE-II

1. (1) Orange, Blue and Red are on the adjacent faces of yellow colour. Again, Blue, Orange, Red and Purple colours are on the adjacent faces of violet colour. Therefore, violet colour will be on the face opposite to yellow colour.

2. (4) The six colours are : Indigo, Violet, Orange, Red, Green and Blue.

The colours Indigo, Blue and Red are on the faces adjacent to Green and therefore, these colours cannot be on the face opposite to Green.

After careful observation of all the views of dice, it is clear that Orange lies opposite to Green.

3. (4) From figure 2 and 4 are adjacent to both 3 and 6 hence 6 will be opposite to 3. '3' lies opposite '6'.

4. (3) Clearly, there are six blocks in the diagram.

5. (4) The numbers 1, 3, 4 and 5 are on adjacent faces of the number 6. Therefore, 6 lies opposite 2.

6. (4) There are seven blocks in the given figure.

7. (2) Six cubes are visible and four cubes are invisible. Thus, there are 10 cubes.

8. (3) 2, 4, 1 and 3 are adjacent to 5. So 6 will be opposite to 5. Now keeping position of 5 stable in both figures respectively. 4, 2 and 3 are in same order, hence 3 will be opposite to 4.

9. (2) From the two views of cube it is clear that 2 dots are contained on the face opposite to that containing 3 dots.

10. (2) The central cube of each face will have only one face painted. Thus, there are six such cubes.

11. (2) Clearly, concentric circles will be on the face opposite to face having symbol star.

12. (1) The numbers 2, 4, 5 and 6 are on the adjacent faces of the number 3. Therefore, 1 lies opposite 3.

13. (2) From Ist and IInd figures the numbers 1, 2, 5 and 6 cannot be on the face opposite the number 4. Therefore, 3 lies opposite 4.

14. (4) When the question figure is folded into a box it will look like answer figure (4).

15. (2) From the two views of the dice, it is clear that 2 lies opposite 4.

16. (4) Clearly, the similar figures will be on the opposite faces of one another.

17. (1) The invisible number is 4.

The numbers 1, 3, 5 and 6 are on faces adjacent to 2.

Therefore, 4 lies opposite to 2.

18. (4) From the four views of the cube it is clear that

4 lies opposite 6.

3 lies opposite 1.

2 lies opposite 5.

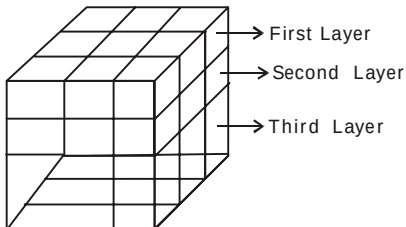
Therefore, the number 3 will be in the place of question mark.

19. (1) One, two, five and six dot(s) are on the faces adjacent to the face having three dots. Therefore, there should be four dots on the face opposite to face having three dots.

| CUBES AND DICE |

Two, three, four or five dots cannot be on the face opposite to face having six dots. Therefore, one dot lies opposite the face having six dots.

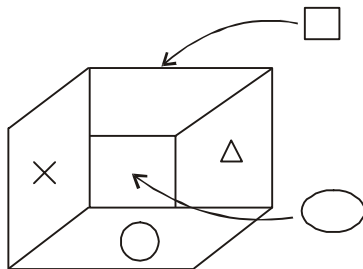
20. (2)



4 cubes each of the first and third layers will have paint on two sides only.

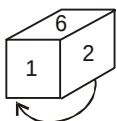
Therefore, total number of cubes having paint on two sides
 $= 4 \times 2 = 8$

21. (4) According to question



The circle as at the bottom.

22. (3) From the two views of cube it is clear that '3' lies opposite to '1'.



23. (2) If there is one triangle at the bottom, there would be three triangles on the top.

24. (2)

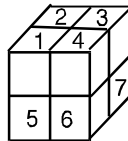


25. (3) The six symbols on the six faces of the given cube are : O, Δ, +, -, × and ÷. From figure 1,2 and 4, ÷, ×, - and + can not be opposite to circle.

Hence, from the four positions of the cube it is clear that the circle lies opposite triangle.

26. (2) The symbols ÷, Δ and O lie on the adjacent faces of the symbol + and hence these cannot be on the opposite face of +.
 Considering all the four views of dice, it is clear that
 Δ lies opposite O.
 + lies opposite ×.
 ÷ lies opposite -.

27. (2)



One block is completely hidden from the view.

28. (3) From the two views of the dice it is clear that dice has been inverted as three dots are on the same side.

Therefore, when two is at the bottom, four will be on the top.

29. (1) The numbers 1, 2, 3 and 4 cannot be at the bottom when 5 is on the top. Therefore, 6 will be at the bottom.

30. (2) On dividing in 27 parts there will be 3 parts of each edge. One cube of each face will be coloured on only one face.

Therefore, $6 \times 1 = 6$

31. (3) Number of blocks
 $= 1 + 4 + 2 = 7$

32. (2) Six cubes are visible.

Hidden cubes = $2 + 1 + 1$
 Total number of cubes = 10

33. (2) From the diagram it is clear that there are 14 blocks.

34. (1) Five will be opposite to Four.
 One will be opposite to Three.
 Two will be opposite to Six.

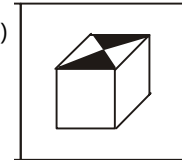
35. (4) The four triangular portions will combine to form a face of the type  which lies opposite to the face bearing the circle.

36. (3)



37. (2) Stem will be opposite of Leaf.
 Bud will be opposite of Flower.
 Fruit will be opposite of Seed.

38. (4)



39. (2) The colours yellow, green, blue and red are on the adjacent faces of purple colour. Therefore, orange lies on the face opposite to purple.

40. (4) From first, second and third figure Orange, Red, Silver and White cannot be on the opposite face of Green colour. Therefore, Violet is opposite to Green.

41. (3) 1, 2, 3 and 5 are adjacent to 4. So 6 will be opposite to 4.

42. (2) Number of cubes having two colours = On every surface $4 \times \frac{6}{2}$

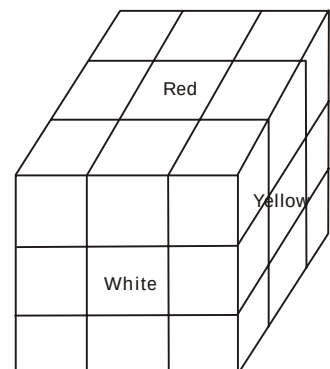
surfaces = $4 \times 3 = 12$



43. (2) The central cube of middle row will be without paint.

44. (1) In 24 small cubes two sides would be seen.

45. (1)

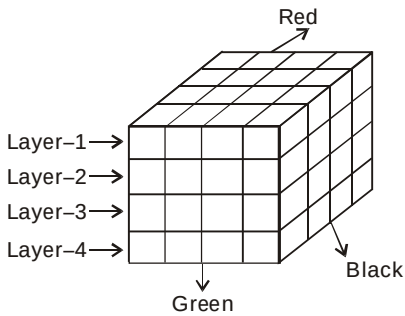


The cubes of middle row will have no red colour $\Rightarrow$ 9 Cubes

The Central cube will have no colour

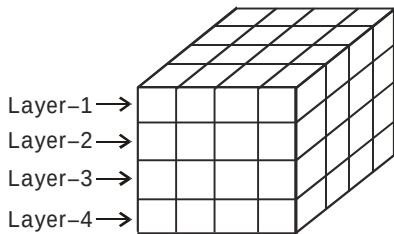
Now, out of 8 cubs, 4 cubes have either yellow or white colour.

46. (2)



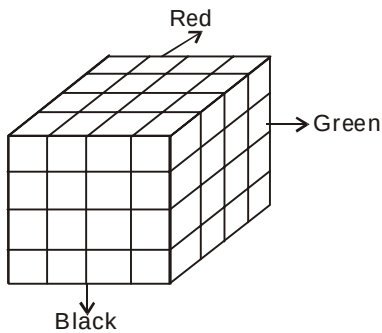
There are four cubes in Layer-I and four cubes in Layer IV which have only one face painted red and all other faces not painted at all. Thus there are eight such cubes.

47. (2)



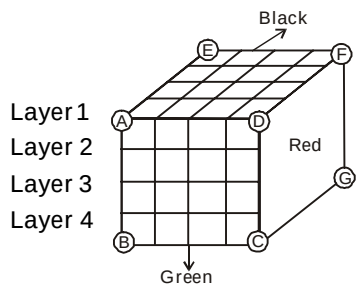
In layers I and IV four cubes each at the four corners will have three faces painted.

48. (1)



No cube will have four faces painted.

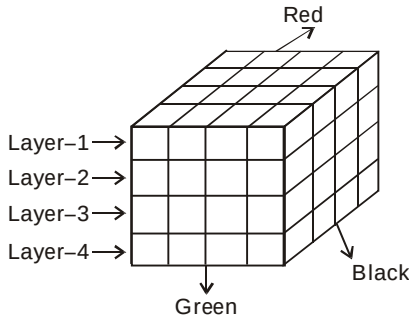
49. (3) Cubes A to G = 7 and 1 backside cube will have 3 colours



Alternatively,

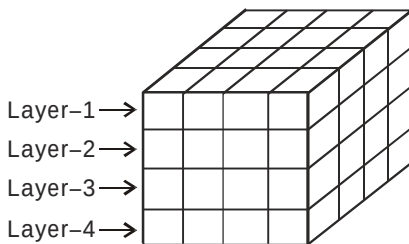
The four cubes at the corners of Layer 1 and the four cubes at corners of Layer 4 have three colours.

50. (3)



There are $4 + 4 = 8$ cubes in layer -1 and $4 + 4 = 8$ cubes in Layer -4 which have one face painted green and one face painted red. Thus there are 16 such cubes.

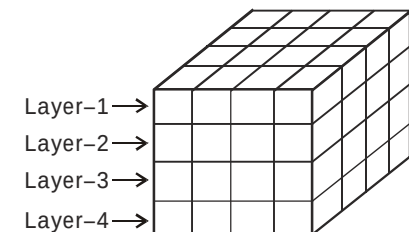
51. (3)



There are 8 cubes in Layer 1, 4 cubes in Layer 2, 4 cubes in Layer 3, and 8 cubes in Layer 4 which have only two faces painted.

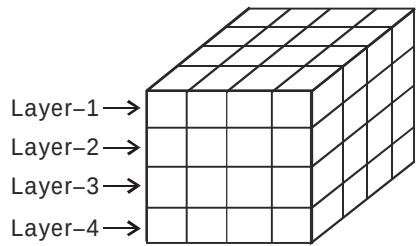
Thus, there are $8 + 4 + 4 + 8 = 24$ such cubes.

52. (3)



Four central cubes in Layer 2 and four central cubes in Layer 3 have no face painted. Thus, there are 8 such cubes.

53. (4)

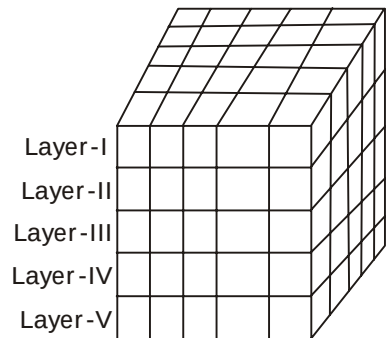


In Layers 1 and 4 the four cubes each in the centre have only one face painted. In Layers 2 and 3, eight cubes each in the centre have one face painted

Thus, total number of cubes
 $= 8 + 16 = 24$

54. (4) There are 12 cubes.

55. (3)



In Layer-I, the nine central cubes have only one face painted, four cubes at the corner have three faces painted and the remaining 12 cubes have two faces painted.

In each of the Layer-II, III and IV, the nine central cubes have no face painted, the four cubes at the corner have two faces painted and the remaining 12 cubes have one face painted.

In Layer-V, the nine central cubes have only one face painted, the four cubes at the corner have three faces painted and the remaining 12 cubes have two faces painted.

Thus, the number of cubes having three faces painted
 $= 4 + 4 = 8$

56. (3) Total number of cubes
 $= 6 \times 6 \times 4 = 144$

57. (1) Altogether there are $9 + 5 + 1 = 15$ cubes. 10 cubes are seen in the figure. So, five cubes are unseen in the figure.

58. (4) No Cube is there with two red faces only. All the eight cubes have three red faces.

59. (4) If we fold the option (1) the number 2 will lie opposite 5.

If we fold the option (2) the number 1 will lie opposite 3.

If we fold the option (3) the number 2 will lie opposite 5.

Therefore, Answer Figure (4) is correct.

60. (2) When folded in the form of a cube, then 'F' appears opposite 'B'; 'E' appears opposite 'C' and 'A' appears opposite 'D'.

In option (1) 'F' is adjacent to 'B'.

In option (3) 'E' is adjacent to 'C'.

In option (4) 'D' is adjacent to 'A'.

61. (2) From the two views of blocks it is clear that when 10 is at the bottom, number 12 will be at the top.

62. (1) 10 cubes are visible and 10 cubes are hidden. Clearly, there is one column having four cubes. There are two columns each having three cubes.

There are three columns, each having two cubes.

There are four columns, each having only one cube.

Thus, total number of cubes
 $= 4 + 6 + 6 + 4 = 20$ cubes

63. (2) The numbers 2, 4, 5 and 6 are on the adjacent faces of number 3. So, number 1 lies opposite 3.

The number 5 is on the adjacent faces of numbers 4 and 6. So, number 5 lies opposite 2.

64. (3) 

65. (2) Three cubes each in the column I and column II from the left are not seen. Similarly, two cubes each in column III and IV are not seen.

Total number of unseen cubes
 $= 3 + 3 + 2 + 2 = 10$

66. (3) The seven colours of a Rainbow are : Violet, Indigo, Blue, Green, Yellow, Orange and Red. Thus, Violet colour of rainbow is left out.

67. (3) From the two views of the same dice it is clear that, B, C and E are on the faces adjacent to A. So, D lies opposite to A.

68. (2) Yellow, Green, Red and Blue colours are on the faces adjacent to Purple. So, Orange colour is opposite to Purple.

69. (2) After folding the sheet of paper in the form of a cube :

 will lie opposite Δ

 will lie opposite $\heartsuit$

 will lie opposite $\clubsuit$

Symbols  and $\heartsuit$ cannot be on adjacent faces. So, Answer Figure (3) can be ruled out.

Symbols  and $\clubsuit$ cannot be on adjacent faces. So, Answer Figure (4) can be ruled out.

 and Δ can not be on adjacent sides. Hence options (1) is not true. Hence option (2) is true.

70. (2) The dot will lie opposite one of the shaded surfaces. Therefore, option (2) cannot be formed.

71. (3) When paper is folded in the form of a cube, then

$\triangle$ lies opposite 
 $+$ lies opposite $\div$
 lies opposite 

72. (3) The shape given in the option (3) is not appropriate.

73. (3) When block is made

$*$ lies opposite
 lies opposite
 lies opposite

74. (2) Red, Orange, Yellow and Green colours are on the faces adjacent to white colour. Therefore, Blue colour lies opposite white colour.

75. (2) Symbols $\triangle$, $\diamond$, $\square$ and $\triangleleft$ lie on the faces adjacent to the symbol . Therefore, $\odot$ lies opposite .

76. (3) The numbers 2, 3, 4 and 5 lie on the faces adjacent to the number 6. Therefore, when top is 6, the number 1 is at the bottom.

77. (2) $*$ lies opposite .

 lies opposite $\boxtimes$.

$\uparrow$ lies opposite Δ .

78. (1) Six cubes are visible and four cubes are hidden.

79. (1) Y, Z, A and C are on the faces adjacent to X.

Therefore, B lies opposite X.

X, Y and A are on the faces adjacent to C. Therefore, Z lies opposite C.

80. (3) Red, Blue, Yellow and White colours are on the adjacent faces of Green colour. Therefore, Orange lies opposite Green colour.

81. (4) Yellow, Green, Blue and Violet are on the faces adjacent to Red colour. Therefore, sixth colour is opposite to Red.

From the four views of cube, it is clear that yellow is opposite to Blue.

82. (2) A, C, D and F are on the faces adjacent to face B. Therefore, E lies opposite B.

B, C, E and F are on the faces adjacent to face A. Therefore, D lies opposite A.

83. (3) Red, Blue, Green and Orange colours are on the faces adjacent to Purple colour. Therefore, yellow colour is opposite to purple colour.

84. (3) Orange, Yellow, Red and Black colours are on the faces adjacent to Pink colour. Therefore, Blue colour is opposite to Pink colour.

Pink, Black, Blue and Yellow colours on the faces adjacent to Red colour. Therefore, Orange colour is opposite to Red colour.

85. (1) Orange, Yellow, Black and Pink colours are on the faces adjacent to Red colour. Therefore, Blue colour is on the face opposite to Red colour.